

Reproducing the LHS 1140 b helium line with a solved wind: how far EXHALE gets, and what is missing

1. Goal, and the answer up front

The objective is to reproduce the measured He I 10830 Å transit line of LHS 1140 b with EXHALE, which solves the wind rather than imposing it, and to establish what — if anything — has to be added to the physics for the model to match the observation.

Result. The *amount* of absorption can be matched, but the composition it takes is not one number: it splits in two according to what is placed at the lower boundary. Above a prescribed metal-free scalar base the measured equivalent width is reached at He/H = 2.09; above a lower atmosphere solved as a photochemical column and handed over as a profile, C, N and O included, it is reached at He/H = 10.7. Both runs have binary H/He element diffusion active and both sit at the same $K_{zz} = 10^9 \text{ cm}^2 \text{ s}^{-1}$, so what separates the two values is the base and not the mixing. The *width* is not matched over either base, and the velocity it asks for is the same number every time. Solving the lower atmosphere instead of prescribing it does not remove the match — it moves it.

(i) The amount is always matchable, so what the line returns is a conditional pair rather than a measurement.

The prescribed-base branch. With the base a scalar state carrying no metals, every combination of assumed spectrum and assumed mixing reaches the observed red-pair equivalent width at some helium abundance, and the conditions stack inside this branch. In the well-mixed limit the crossing is He/H = 0.55 on the GJ 1132 proxy spectrum and 0.060, below solar, on the GJ 699 one — the second stand-in for LHS 1140's unobserved XUV, 5.1 times brighter, which the paper also uses. With binary H/He element diffusion active — helium then settles out of the base unless an eddy term lifts it — it rises again, to 4.68 wherever $K_{zz} \leq 10^5 \text{ cm}^2 \text{ s}^{-1}$ does nothing against the molecular coefficient, 2.09 at the adopted $K_{zz} = 10^9$ and 1.11 at 10^{11} (Sect. 8.3). The dependence on the eddy coefficient is weak and monotonic, as $K_{zz}^{-0.14}$ wherever the eddy term acts at all; the dependence on the assumed spectrum is a factor 9.

The solved-column branch. Replacing that base with a photochemical column — Photochem with its climate step solved, handed to EXHALE as a profile at a 1 μbar match, at the same $K_{zz} = 10^9$ and with the same diffusion active — brings C, N and O into the base, and the same reservoir then absorbs $2.6\times$ less: He/H = 2.09 gives $EW = 0.421 \% \text{ Å}$ against the observed 1.108 ± 0.030 , where over the metal-free base it gave 1.129 (Sect. 11.5). A ladder of seven solutions on this base, each closed on its own elemental fluxes, crosses the measurement at He/H = 10.7 (10.5–10.9 on the measurement's 1σ interval), read off the bracket between the 10.31 and 12.01 rungs (Sect. 11.6).

Neither number is an independent prediction of the model. Each is the reservoir the observed line picks out over its own base, and the escape rate — which is an output, not a fit — barely moves along the way. The flux closure does not change that on the profile branch. It returns He/H = 2.09235 at the match from trial fluxes a factor of ten apart, single-valued to 5.5×10^{-6} (Sect. 11.), but at $K_{zz} = 10^9$ the eddy term homogenizes the column all the way to the microbar match, so only 2×10^{-4} of that answer comes from the escape flux: the closure hands back the reservoir it was given. And the reservoir it was given, 2.09, is itself the value fitted to this same line over the earlier metal-free scalar base. What the closure adds is that a composition is consistent with a photochemical lower atmosphere carrying its own C,

N and O, not that it is measured. Every rung of the ladder satisfies the closure at residual flux spreads of 0.4–2.6%; the line is what selects one.

Why the same helium absorbs differently over the two bases. The base metals are a thermostat setting, and helium is the other half of the same thermostat. The photochemical column's C, N and O cool the wind — $T_{\max}$ from 5330 to 3944 K — and although the metastable peak itself rises, 42.9 to 50.2 cm⁻³, the cooler wind does not carry the population outward, so the metastable column above $2R_p$ falls by 2.6 and the line weakens by the same factor. Raising the reservoir reheats the wind: at He/H = 10.31 the mean mass per particle at $2R_p$ is $2.53m_H$ against 1.51, the same deposited energy is shared among fewer and heavier particles, the coolant is diluted with the hydrogen it is referred to, and $T_{\max}$ returns to 5869 K — past the metal-free 5330. The ladder is therefore closer to a thermal trade between base metal content and helium content than to a determination of composition, and this one line cannot separate the two (Sect. 11.5, Fig. 16).

The escape rate is flat on every axis, and end to end the composition spans a factor 180. $\log_{10}(\dot{M}/\text{g s}^{-1}) \simeq 7.6\text{--}7.8$ on the GJ 1132 proxy across composition, diffusion and eddy coefficient alike, 7.39–7.50 across the whole closed ladder, and 8.6 on the GJ 699 proxy — there within 0.04 dex of what the paper itself retrieves on that same proxy (Sect. 10.). Against that, the composition the line implies runs from 0.060 to 10.7 over choices this measurement cannot arbitrate: strongly conditional on the assumed spectrum and on what is placed at the lower boundary, weakly and monotonically on the assumed mixing. Any composition read off this line — the paper's H:He $\lesssim 10^{-3}$ as much as any value here — inherits those conditions.

(ii) The width is not reached on any axis, and the deficit is invariant. At the equivalent-width crossing the profile is about 3.4 times too deep and 3.4 times too narrow, and no composition fixes it, because composition changes how much helium absorbs, not how the absorption is spread in velocity. Measured directly — convolving the solved profile with a Gaussian and asking which one reproduces the observed line — the missing spread is 22.30 km s⁻¹ FWHM ($\sigma = 9.5$ km s⁻¹) in the well-mixed limit, 22.31 km s⁻¹ with diffusion and the adopted K_{zz} , 22.39 km s⁻¹ for the flux-closed solution on the solved photochemical column, and 22.08 km s⁻¹ on the $5.1\times$ brighter spectrum (Sects. 8.2, 8.3, 11. and 10.). Supplied as one Gaussian it puts depth, blue depth and red/blue ratio inside the observed 1σ interval at once, with nothing tuned to them; what produces it is unidentified. It is not thermal — the solved gas cools below 1000 K over the line-forming region — and not bulk outflow, which reaches 1.2 km s⁻¹ at $20R_p$ on the fainter spectrum and 3.6 on the brighter. Read as turbulence it is $\mathcal{M} = 2.4\text{--}4.2$ against the sound speed these solutions themselves carry, supersonic on every convention and well above what the literature is willing to assume. The one mechanism that would supply width without a free parameter — the adiabatic-cooling metastable bump of Schulik & Owen (2025) — is present in these solutions and measured, but sits at 0.3–1.4 km s⁻¹ (Sect. 8.5). The shortfall is in the energy and momentum budget, not in the composition, and it is a published problem rather than a defect of this setup (Sect. 8.). The paper's own p-winds fit does not reproduce the line either: it carries roughly the right width, because its temperature is imposed at 5000–6000 K where the line forms, but overshoots the blue component threefold. Adding broadening does not rescue it: its absorbing power is 24% below the measurement, so no convolution can reach the observed depth and width at once.

A caveat that attaches to the escape rate alone. These solutions stay subsonic out to $30R_p$, so their mass flux is set at

the outer boundary rather than at a critical point, and the gas is already collisionless there ($\text{Kn} = 1.3\text{--}2.3$): the $\dot{M}$ quoted throughout is *unvalidated in the collisional sense*, while the He 10830 layer at $1.1\text{--}3 R_p$ sits at $\text{Kn} \leq 0.021$ and the line analysis is untouched (Sect. 12.).

Everything below uses inputs verified against the authors' own released material (Zenodo 20723095): their p-winds driver script, their Mega-MUSCLES v23 spectra, their reduced transmission spectrum and their likelihood grid. Products live in LHS1140b/; the notebook LHS1140b/LHS1140b_analysis.ipynb regenerates every number and figure here.

2. Inputs, and how each was pinned

2.1 System

Planet and star from Cadieux et al. (2024) as quoted by C26: $R_p = 1.730 \pm 0.025 R_\oplus$, $M_p = 5.60 \pm 0.19 M_\oplus$, $T_{\text{eq}} = 226 \text{ K}$, $a = 0.0946 \text{ AU}$, $M_\star = 0.1844 M_\odot$, $R_\star = 0.2159 R_\odot$. One trap: EXHALE converts R_p with its own internal constants (mean radii, $R_J = 6.9911 \times 10^9 \text{ cm}$, $R_\oplus = 6.3725 \times 10^8 \text{ cm}$), so the input must be built with those and not with IAU equatorial values — a 2.2% error in radius, 4.4% in transit depth, if taken from the wrong pair.

2.2 Lower boundary condition

EXHALE anchors the lower boundary at the planetary radius R_p using the standard CETIMB pressure-anchored condition (Base BC: pressure 1.0), fixing the base pressure at $P_0 = 1.0 \mu\text{bar} = 1.0 \text{ dyn cm}^{-2} = 1.0 \text{ erg cm}^{-3}$ (0.1 Pa). The corresponding base total number density n_0 follows from the ideal gas law at the equilibrium temperature $T_0 = T_{\text{eq}} = 226.0 \text{ K}$:

$$n_0 = \frac{P_0}{k_B T_0 n_{\text{tot, bc}}} = \frac{1.0 \text{ dyn cm}^{-2}}{(1.380649 \times 10^{-16} \text{ erg K}^{-1}) \times 226.0 \text{ K}} = 3.20485 \times 10^{13} \text{ cm}^{-3}, \quad (1)$$

where $n_{\text{tot, bc}} = 1.0$ for neutral atomic gas. In logarithmic form, $\log_{10}(n_0/\text{cm}^{-3}) = 13.506$, which is entered in `input.inp` as `Log10 lower boundary number density [cm^-3]: 13.506`. When Base BC: pressure is specified, EXHALE dynamically re-evaluates and enforces this relation internally (`input_read.f90`).

2.3 Stellar spectrum

The paper describes its normalization two ways that cannot both hold: the Supplement's prose says the proxy SED is scaled so its integrated X-ray flux matches that measured for LHS 1140, while the quoted constants ($A_{1132} = 0.59$, $A_{699} = 0.056$) are the catalog X-ray flux *ratios*, a factor 1.8 apart. The authors' script settles it:

```
Lx_ratio_1132 = 0.59
```

```
norm_constant_1132 = Lx_ratio_1132*(d_1140/(smax_1140*au2m))**2
```

applied to the plain `const-res v23` product. That combination also reproduces the fiducial XUV flux the main text quotes, 0.033 W m^{-2} (we measure $33.6 \text{ erg s}^{-1} \text{ cm}^{-2}$ over $10\text{--}1300 \text{ \AA}$). All results here use it. Note the proxy SED and the paper's own APEC fit disagree in the X-ray band (4.4 against $2.9 \text{ erg s}^{-1} \text{ cm}^{-2}$ at the orbit); that inconsistency is inherited from the paper's procedure, not introduced here.

2.4 Measurement

The comparison target is the authors’ released Fig. 3B spectrum (129 samples, vacuum, planetary rest frame, correlated-noise corrected), not a reading of a figure. Fitting it with our own three-Gaussian extractor (`he_line_metrics.py`, the same one applied to every model spectrum here) reproduces the paper’s MCMC to within its published uncertainties:

	this extractor	paper
blended red depth	1.254%	$1.24^{+0.22}_{-0.23}\%$
blue depth	0.198%	$0.25^{+0.14}_{-0.12}\%$
red/blue	6.35	$6.7^{+12.7}_{-3.1}$
FWHM	0.841 Å	$0.86^{+0.15}_{-0.27}$ Å

so models and measurement can be compared through one instrument.

3. Reproducing the paper’s p-winds model

Run at the authors’ settings — their SED normalization, $R_p = 0.154 R_J$, $b = 0.23$, `turbulence_broadening=True`, five-phase transit averaging, and their fitted line-of-sight bulk velocity $v_{\text{wind}} = 2.26 \text{ km s}^{-1}$ — and at the MCMC medians ($\dot{M} = 2.03 \times 10^8 \text{ g s}^{-1}$, $T = 5160 \text{ K}$, $\text{H:He} = 1.01 \times 10^{-3}$), the vector their Fig. 4 legend states for the plotted curve, our reproduction gives a red depth of 1.376% and FWHM 0.575 Å against the 1.32% and ~ 0.71 Å read off their Fig. 4: a 3% match in depth, but a line about 20% too narrow.

The width residual is not a consequence of which point in parameter space is drawn. Running the same reproduction at the grid maximum-likelihood point ($1.83 \times 10^8 \text{ g s}^{-1}$, 5131.6 K, 8.338×10^{-4}) gives red 1.366% and FWHM 0.581 Å, i.e. the two vectors differ by less than 0.01 Å in width. What produces the extra width in the published curve has not been identified.

Two points are worth recording because both cost time to find.

First, the curves drawn over the spectrum in the paper’s Fig. 3 are the *three-Gaussian empirical fits*, not p-winds; the physical model is overlaid only in Fig. 4. Fig. 3’s apparently perfect agreement is a fit to the data, and says nothing about the model’s line shape.

Second, the paper’s plotted model is visibly broader than a plain p-winds run, which we initially modelled as an undocumented instrumental kernel. It is not: their script sets `turbulence_broadening=True` and averages over five transit phases. Both widen the line, and together they account for the difference.

4. Turbulent broadening: what the literature does

Since that switch turned out to matter, all 43 papers under references/ were checked: every PDF converted to text, searched for `turbulen`, `microturbulen` and `non-thermal broadening`, and each hit read in context. Fifteen matched; most use “turbulent” for eddy diffusion K_{zz} or for flow structure in 3D simulations, not for line broadening. Section 4.1 tabulates the ones that bear on broadening.

The convention, stated most explicitly by Yan et al. (2024), replaces the thermal Doppler width by

$$\Delta\nu_D = \frac{\nu_0}{c} \left(\frac{2kT}{m} + v_{\text{turb}}^2 \right)^{1/2}, \quad v_{\text{turb}} = \sqrt{\frac{5kT}{3m}} = c_s, \quad (2)$$

the monatomic sound speed, as assumed by Salz et al. (2018), Lampón et al. (2020) and Czesla et al. (2022). The p-winds implementation uses $\sqrt{5kT/6m}$ in a σ -convention, which is the same physics expressed against a Doppler width that lacks the factor 2; in EXHALE’s b -parameter convention the same term is $b \rightarrow \sqrt{11/6} v_{\text{th}}$, a factor 1.354. That is what EXHALE_TRANSIT_TURB=1 applies. It is opt-in and off by default, because the field does not agree that it belongs.

4.1 Paper by paper

Adopt it as an assumed value. Salz et al. (2016), Eq. (5): “ b is the Doppler parameter, v_{th} the thermal Doppler velocity, and for the microturbulence v_{micro} we use the sound speed”, applied when synthesizing the transit profile. (Their atmosphere solver separately carries a constant 1 km s^{-1} microturbulence at the lower boundary, a different quantity, which they note “has little impact on the overall atmospheric structure”.) Lampón et al. (2020, 2021) is the reference implementation, with turbulent broadening as a free or assumed parameter of 1D isothermal Parker fits; Czesla et al. (2022) follows it. Cherubim et al. (2026) inherits the choice through p-winds’ turbulence_broadening=True, so the LHS 1140 b retrieval carries it — a fact stated nowhere in the paper text.

Test it, then leave it out. Yan et al. (2024) give the cleanest statement of the convention and of the doubt. They record that “these authors assumed that the turbulence velocity $v_{\text{turb}} = \sqrt{5kT/3m}$, which corresponds to the speed of sound of a monatomic gas and, thus, to the transonic case with a Mach number of 1”, then keep it out of their nominal models and test $v_{\text{turb}} = 0.5, 1.0$ and $1.5 c_s$ as a sensitivity. Their finding for He 10830 is that turbulence “decreases the absorption depth slightly at the line center, but doesn’t significantly alter the absorption at the line wings”. They argue Mach 1 is likely an overestimate, since in the diffuse ISM turbulent velocity anti-correlates with temperature and the Mach number is below unity at a few thousand K, and conclude “the turbulence effect is likely less significant than the one explored in this paper”. Yan et al. (2019) computes the Doppler width explicitly “assuming no turbulence”.

Argue against it. Murray-Clay et al. (2009), on García Muñoz’s (2007) proposal to broaden $\text{Ly}\alpha$ by wind turbulence: “to generate the large velocities observed, the energy in such turbulence would need to exceed the thermal and bulk kinetic energies in the mean flow by a factor of ~ 100 . Such energy requirements seem insurmountable.” Koskinen et al. (2013b) find that “non-thermal broadening such as that proposed by Ben-Jaffel and Hosseini (2010) does not appear to be necessary to explain the current observations”, while allowing that stellar-wind interaction may produce broadening turbulence. Schulik & Owen (2025) offer a physical alternative — adiabatic cooling produces a metastable-helium population bump at high velocity, widening the line with no free parameter — and are pointed about the alternative: the isothermal model “can only explain the broadness of the data under physically inconsistent assumptions, i.e. those of even larger mass-loss rates, not supported by the energy input into this planet, and an empirical free parameter, turbulent broadening (Lampón et al. 2021)”. They add that $T = 10^4 \text{ K}$ already maximizes what thermal broadening alone can give.

Need it even so. Taylor et al. (2026) is the closest analogue to the present work: a self-consistent model (energy balance, multispecies diffusion, non-LTE $H(n=2)$) that reproduces the HD 189733 b line core but not its width. They write that “several prior analyses have matched the HD 189733 b width by introducing ad hoc (micro)turbulent or Gaussian broadening terms; here, we show that this step remains necessary even in our self-consistent framework”, and quantify it as $\sim 12 \text{ km s}^{-1}$ of additional non-thermal broadening. They speculate — explicitly calling it speculative — that it is plasma turbulence from ions trapped on low-latitude planetary field lines, strengthening with stellar activity. Rumenskikh et al. (2022), as reported there, instead imposed reduced hydrogen line cooling and prescribed zonal jets of $10\text{--}20 \text{ km s}^{-1}$ in a 3D multifluid model.

Matches that are not about broadening. Recorded so the sweep need not be repeated: Wogan (2025), Lavvas (2014, 2019), Robeling (2026) and the second half of Koskinen et al. (2013b) use “turbulent” for eddy diffusion K_{zz} ; 2607.18193v1 for flow–flow interaction in a 3D stellar-wind simulation; Jensen (2012) for photospheric turbulence driving stellar activity; Taylor et al. (2025) for gravity-wave breaking. Vidal_2003_Nature.pdf matches only on its first page, which carries the tail of the preceding *Nature* letter (on Crab pulsar radio emission); Vidal-Madjar et al. themselves do not discuss turbulence.

4.2 Empirical baseline: turbulence in Solar System giant planets

To evaluate whether sonic or supersonic ($v_{\text{turb}} \sim c_s$) turbulence is physically plausible in a planetary atmosphere, it is instructive to compare with directly measured turbulence and non-thermal velocity fields across the stratified layers of Jupiter and Saturn:

- *Troposphere and cloud deck* ($\sim 0.1\text{--}1 \text{ bar}$). Cloud-tracking measurements from *Voyager*, *Galileo*, *Cassini*, and *Juno* (e.g., Salyk et al. 2006; Del Genio et al. 2007; Ingersoll et al.) isolate the fluctuating eddy velocities $v' = \sqrt{u'^2 + v'^2}$ from the mean zonal jets ($100\text{--}500 \text{ m s}^{-1}$). The resulting RMS turbulent eddy velocity is $v' \approx 3\text{--}10 \text{ m s}^{-1}$ on Jupiter (locally $20\text{--}30 \text{ m s}^{-1}$ in the Great Red Spot shear zone) and $5\text{--}15 \text{ m s}^{-1}$ on Saturn. Compared to the tropospheric sound speed ($c_s \approx 800\text{--}900 \text{ m s}^{-1}$), this is deeply subsonic turbulence, with Mach numbers $\mathcal{M} \sim 0.005\text{--}0.02$.
- *Stratosphere* ($\sim 10 \mu\text{bar}\text{--}10 \text{ mbar}$). High-resolution submillimeter spectroscopy of impact-delivered HCN/CO molecules with ALMA (e.g., Cavalié et al. 2021) and Herschel reveals stratospheric jet velocities up to $300\text{--}400 \text{ m s}^{-1}$ in the auroral zone, but the associated non-thermal velocity dispersion (turbulence and wave motions) remains limited to tens of m s^{-1} up to $\sim 100\text{--}200 \text{ m s}^{-1}$ ($\mathcal{M} \lesssim 0.2$).
- *Thermosphere and auroral ionosphere* ($\lesssim 1 \mu\text{bar}$). Ground-based high-resolution ($R \sim 10^5$) infrared spectroscopy of H_3^+ line profiles (e.g., with VLT/CRIRES, Keck, and CFHT) disentangles thermal Doppler broadening from non-thermal velocity fields. In non-auroral mid-latitudes, the non-thermal velocity is $v_{\text{nth}} \approx 0.2\text{--}0.5 \text{ km s}^{-1}$. In the auroral oval, where intense magnetosphere–ionosphere coupling, ion drag, and Joule heating inject substantial kinetic energy, non-thermal broadening reaches $v_{\text{nth}} \approx 0.8\text{--}2.0 \text{ km s}^{-1}$.

These empirical benchmarks show that pure hydrodynamic planetary turbulence is strictly subsonic ($\mathcal{M} \ll 1$), and even violent magnetospheric auroral driving in upper thermospheres caps non-thermal broadening at $\lesssim 2 \text{ km s}^{-1}$.

4.3 Turbulence and 3D hydrodynamic simulations of exoplanets

Beyond 1D parameterizations, 3D numerical simulations explore how turbulent mixing, non-thermal velocity structures, and line broadening emerge across diverse exoplanetary regimes:

1. 3D hot Jupiter escape and circulation.

- *3D atmospheric escape and kinematic line broadening.* Hydrodynamic 3D escape simulations (e.g., McCann et al. 2019; Debrecht et al. 2020; Carolan et al. 2020, 2021; Wang & Dai 2021; MacLeod & Oklopčić 2022) demonstrate that apparent transmission line broadening arises primarily from large-scale 3D kinematic velocity gradients ($10\text{--}50\text{ km s}^{-1}$) across the day-to-night outflow, Coriolis-deflected tails, and tidal distortion, rather than unresolved isotropic microscale turbulence. Furthermore, the collision between the expanding planetary wind and the incoming stellar wind creates a shear layer prone to Kelvin–Helmholtz instabilities and turbulent mixing at the contact discontinuity. In 3D multifluid models of HD 189733 b, Rumenskikh et al. (2022, 2023) show that upper-atmospheric zonal jets of $10\text{--}20\text{ km s}^{-1}$ combined with 3D outflow geometry reproduce the observed broad transmission profiles without invoking ad hoc microturbulence.
- *3D general circulation models (GCMs) and macroturbulent mixing.* In the observable lower-to-middle atmosphere (1 bar to $\sim 1\text{ }\mu\text{bar}$), 3D GCMs (e.g., Showman & Polvani 2011; Showman et al. 2013, 2015; Parmentier et al. 2013; Charnay et al. 2015; Zhang & Showman 2018) demonstrate that equatorial superrotation ($1\text{--}5\text{ km s}^{-1}$) and vertical mixing ($K_{zz} \sim 10^7\text{--}10^{11}\text{ cm}^2\text{ s}^{-1}$) are driven by planetary-scale macroturbulence (non-linear interactions between planetary-scale Kelvin and Rossby waves and the mean flow) and overturning circulation. The sub-grid parameterization of small-scale turbulent dissipation (hyperdiffusion, Smagorinsky closures, or Shapiro filters) remains a known source of numerical sensitivity in GCM wind speeds and jet stability (Heng et al. 2011; Polichtchouk et al. 2014; Skinner & Cho 2021).
- *Magnetohydrodynamic (MHD) turbulence in ultra-hot atmospheres.* For planets with $T \gtrsim 1500\text{ K}$, thermal ionization of alkali metals (Na, K) couples the gas to the planetary magnetic field. 3D MHD simulations (e.g., Perna et al. 2010, 2012; Rauscher & Menou 2013; Rogers & Showman 2014; Rogers & Komacek 2014; Beltz et al. 2022) show that magnetic winding of the field by supersonic zonal winds generates Lorentz forces (magnetic drag) that brake the flow, generating 3D MHD turbulence, Ohmic dissipation, and time-dependent variability in wind patterns and hotspot offsets.

2. 3D simulations across other planetary classes. The reach of 3D fluid simulations extends far beyond hot Jupiters to other major categories of exoplanets and substellar objects:

- *Warm Neptunes and sub-Neptunes.* For warm Neptunes such as GJ 436 b and GJ 3470 b, 3D particle-mesh and hydrodynamic escape simulations (Bourrier et al. 2015, 2016; Villarreal D’Angelo et al. 2021; Carolan et al. 2021) reveal gigantic, asymmetric cometary hydrogen tails spanning tens of planetary radii, directly explaining the 56% $\text{Ly}\alpha$ transit obscuration and the He 10830 Å absorption through 3D stellar radiation pressure and charge-exchange interactions. In metal- and water-rich sub-Neptune atmospheres (e.g., GJ 1214 b, K2-18 b), 3D GCMs

model cloud/haze circulation, photochemical transport, and vertical eddy mixing ($K_{zz} \propto P^{-0.4}$; Charnay et al. 2015; Innes & Pierrehumbert 2022).

- *M-dwarf habitable zone terrestrial planets.* Tidally locked rocky worlds (such as TRAPPIST-1 e/f/g, Proxima b, and LHS 1140 b) have been extensively modelled with 3D climate GCMs (Turbet et al. 2016, 2018; Wolf et al. 2017; Sergeev et al. 2020, 2022; the THAI intercomparison, Fauchez et al. 2020, 2022). These simulations demonstrate how thick dayside convective cloud decks elevate planetary albedo to prevent runaway greenhouse states, while stationary Rossby–Kelvin wave patterns redistribute heat to the nightside to avoid atmospheric freeze-out. Complementary 3D multi-species MHD simulations (Dong et al. 2017, 2018; Cohen et al. 2014) assess O^+ and H^+ ion escape driven by extreme stellar coronal mass ejections and flaring winds.
- *Lava worlds and ultra-short-period rocky planets.* For scorched rocky planets with dayside magma oceans (e.g., 55 Cancri e, K2-141 b, CoRoT-7 b), 3D hydrodynamic models of mineral-vapor atmospheres (Castan & Menou 2011; Nguyen et al. 2020; Zilinskas et al. 2022; Kang et al. 2021) reveal supersonic dayside-to-nightside winds ($\mathcal{M} > 1-3$, $v \sim 1-3 \text{ km s}^{-1}$), standing hydraulic jumps/shocks, and mineral vapor condensation leading to rock rain on the nightside.
- *Brown dwarfs and directly imaged giant planets.* In isolated brown dwarfs and young, directly imaged giants (e.g., HR 8799 b/c/d/e, VHS 1256 b), 3D convective-radiative GCMs (Tan & Showman 2019, 2021a,b; Showman & Kaspi 2013; Zhang & Showman 2014) couple rapid planetary rotation (strong Coriolis forces) with deep geothermal convection and silicate/iron cloud radiative feedbacks. The resulting turbulent vortices, patchy cloud bands, and storm dynamics provide the physical explanation for observed rotational photometric variability.

4.4 Where that leaves this comparison

There is no settled treatment. Turbulence at $v_{\text{turb}} = c_s$ is a common convention in 1D isothermal Parker-wind fits — and hence in the paper being reproduced — while Solar System observations and self-consistent or 3D models demonstrate that isotropic microscale turbulence cannot physically reach the required supersonic levels. 3D models instead supply line width through kinematic flow gradients, zonal jets, and magnetospheric/plasma interactions. Enabling the $v_{\text{turb}} = c_s$ term in EXHALE is therefore right for a like-for-like comparison with C26 and wrong as a physical default, which is how it is implemented.

5. Searching for a match: the composition scan

Once the star, the planet and the spectrum are fixed, composition is the free parameter a match has to be sought in first. This section takes it in the well-mixed limit — element diffusion off, so the reservoir ratio is also the ratio everywhere in the wind. That is the simpler of the two mixing assumptions available, and Sect. 8.3 repeats the whole exercise with binary H/He diffusion active, where the answer depends on the eddy coefficient as well.

Eleven compositions were solved, from solar ($\text{He}/\text{H} = 0.083$) to $\text{He}/\text{H} = 10^4$, each a JFNC steady state ($\|R\| < 10^{-3}$, $\text{info} = 0$; one case, $\text{He}/\text{H} = 100$, stagnates at 1.18×10^{-3} and is recorded as such). Nothing is imposed: mass, momentum and energy are solved with photoionization heating and radiative cooling, so $\dot{M}$ and $T(r)$ are outputs.

He/H	$\log_{10} \dot{M}$ [g s ⁻¹]	red [%]	red (turb) [%]	FWHM [Å]	EW [% Å]
0.083	7.77	1.21	1.08	0.218	0.279
0.25	7.75	2.65	2.32	0.230	0.645
0.50	7.76	4.05	3.50	0.242	1.040
0.55	7.76	4.29	3.69	0.244	1.109
0.70	7.78	4.97	4.26	0.248	1.311
1	7.80	5.88	5.01	0.255	1.598
10	7.76	7.42	6.14	0.287	2.270
10 ³	7.74	5.67	4.66	0.304	1.844
10 ⁴	7.73	5.60	4.60	0.304	1.825
observed	—	1.254	—	0.841	1.108 ± 0.030

Nine of the eleven runs are listed; He/H = 0.6 falls between two tabulated entries and He/H = 100 is the case whose residual stagnates. The three helium-dominated rows are corrected values: their depths and equivalent widths were an artifact of the old advection gate, which pinned the metastable population to its equilibrium value across the hydrogen ionization front, and they fell by 89–93% when the gate was rebuilt on the slowest solved species (Update_EXHALE.md Sect. 72; at He/H = 10, red 75.97% → 7.42% and EW 21.53 → 2.270 % Å). The crossing of Sect. 7. is interpolated inside He/H ≤ 1 and is untouched by that correction.

The escape rate is nearly composition-independent, $\dot{M} \simeq 6 \times 10^7 \text{ g s}^{-1}$, about a third of the 2.03×10^8 the retrieval imposes.

6. Why the two models disagree

At the same composition the two models are not small perturbations of each other. The *total* metastable column is comparable — at H:He = 10⁻³ EXHALE’s is about half of p-winds’ — but its distribution is not, and transit depth is absorbing *area*:

	p-winds	EXHALE
50% of the 2 ³ S column inside	1.33 R_p	5.7 R_p
90% inside	1.59 R_p	10.2 R_p

Both radii are read from the local-equilibrium metastable profile, which is what Fig. 2 draws. The advection-corrected profile that the transit integral consumes is more concentrated, and for this helium-dominated composition it collapses onto the base once the advection gate is formed on the metastable’s own clock rather than on hydrogen’s (Update_EXHALE.md Sect. 72). This section is therefore a statement about the two wind solutions; every transit number quoted elsewhere in the memo is measured on the advection-corrected profiles.

The star is 13.6 R_p across, so metastable helium spread to 10 R_p covers most of the disk while a column confined to 1.6 R_p can darken only a narrow ring. The distribution difference is the wind structure — the one thing p-winds imposes

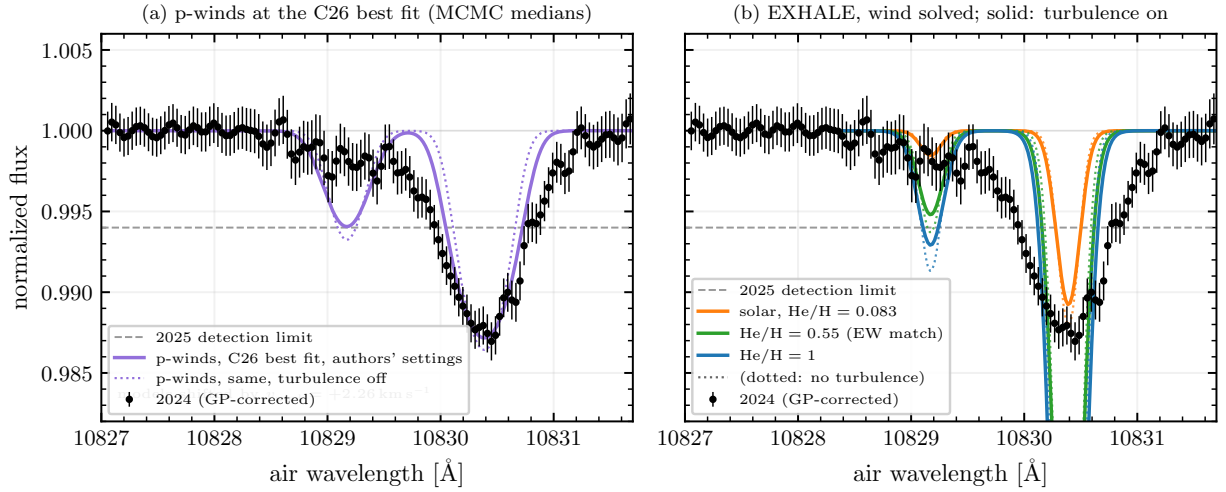


Figure 1: The measurement (black) with, left, the paper’s own p-winds model at its best fit — the MCMC medians its Fig. 4 legend states — computed with the authors’ released settings (solid; dotted is the same run with turbulence off), and right, EXHALE at three compositions. Solid EXHALE curves carry the same turbulence term p-winds uses; dotted are without it. Neither code applies a line-of-sight bulk velocity, so both sets of model curves are shifted by the $v_{\text{wind}} = +2.26 \text{ km s}^{-1}$ that C26 fit; the shift is rigid and leaves equivalent widths unchanged. Drawn in the presentation of the paper’s Fig. 4.

and EXHALE solves. At $r = 10R_p$, same composition and spectrum: p-winds is isothermal at 6100 K by construction, EXHALE’s energy balance gives 1392 K; p-winds flows at 4.3 km s^{-1} , EXHALE at 0.53; and the cold, slow, denser gas keeps helium neutral and the metastable level populated far out, $n(2^3S) = 40 \text{ cm}^{-3}$ against 2×10^{-4} (Fig. 2).

7. What can be matched: the equivalent width

Depth and width trade against each other under any broadening, but the equivalent width does not: it is invariant under the instrument profile, the turbulence term and the phase averaging. It is therefore the quantity on which a self-consistent model can be matched without first settling the broadening question.

Over the blended red pair (vacuum 10832.6–10834.2 Å), the released spectrum gives $EW = 1.108 \pm 0.030 \text{ \% Å}$. EXHALE crosses it at

$$\text{He}/\text{H} \simeq 0.55 \quad (0.53\text{--}0.57 \text{ on the } EW \text{ } 1\sigma), \quad \text{H}:\text{He} \simeq 1.8, \quad (3)$$

that is $6.6\times$ solar, a helium mass fraction of 0.69. Enabling turbulence moves this to 0.541, a 1.6% shift — a direct demonstration that the estimate does not depend on the broadening treatment. The value sits three orders of magnitude away from the paper’s $\text{H}:\text{He} \lesssim 10^{-3}$, and the reason is the structural difference of the previous section: a wind that spreads metastable helium over most of the stellar disk needs far less helium to absorb as much.

Two assumptions are carried inside that number and are taken up later. It is the well-mixed crossing, and it is the crossing on the GJ 1132 proxy spectrum. Sect. 8.3 rescans it with element diffusion active at every decade of the eddy

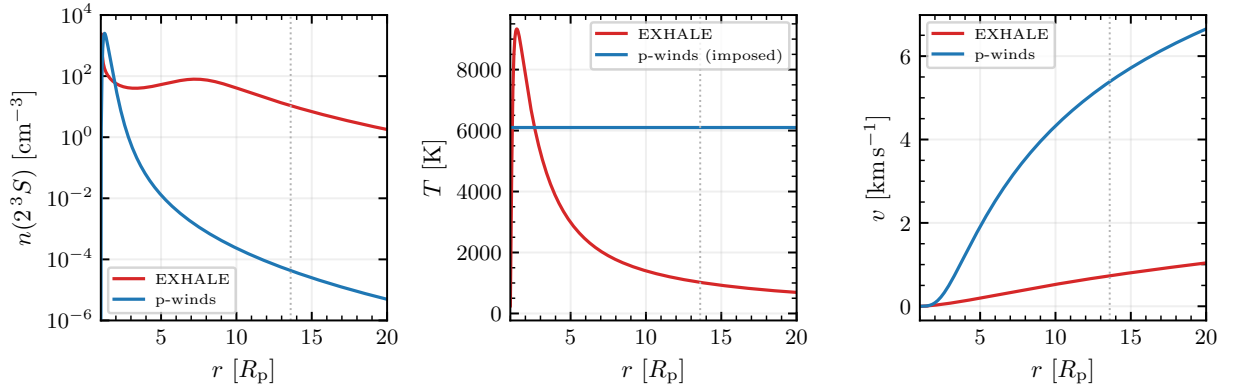


Figure 2: Same composition ($\text{H}:\text{He} = 10^{-3}$), same spectrum, two wind structures. Dotted vertical line: the stellar radius, $13.6R_p$.

coefficient, and Sect. 10. rescans it on the brighter proxy; both move it, the spectrum by much the larger factor.

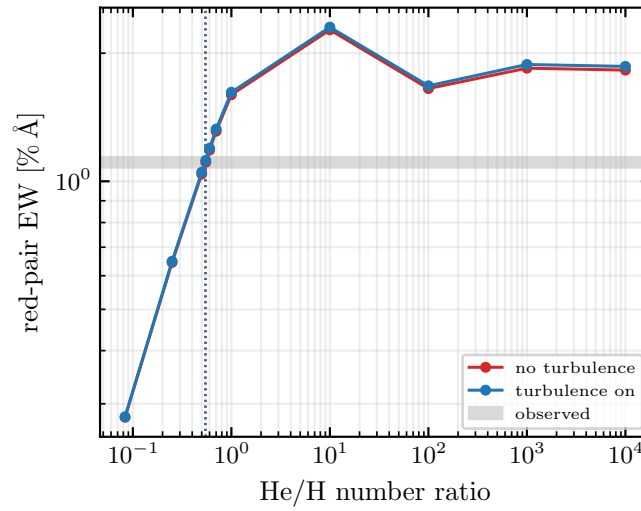


Figure 3: Red-pair equivalent width against composition, with and without the turbulence term, against the measured band. The crossing is insensitive to the broadening treatment.

8. What cannot be matched: the profile

Matching the equivalent width is not matching the line. At $\text{He}/\text{H} = 0.55$ the profile is 3.4 times too deep (4.29% against 1.254%) and 3.4 times too narrow (0.244 against 0.841 Å). The equivalent width states how much absorbing power the model has; the profile states how that power is spread in velocity, and composition does not act on the second. Every solved composition gives 0.22–0.30 Å, rising only to 0.24–0.38 Å with the turbulence term.

8.1 How much is missing, in velocity

The measured FWHM is $0.841 \text{ \AA}$, that is 23.3 km s^{-1} at $10833 \text{ \AA}$. Three ways to supply it, and what each would require:

- *Thermal alone.* A Doppler width of 23.3 km s^{-1} FWHM in helium needs $T = 4.7 \times 10^4 \text{ K}$. Helium is fully ionized far below that, so there would be no metastable population left to absorb. This is excluded.
- *Bulk outflow.* A line-of-sight velocity spread of order $\pm 12 \text{ km s}^{-1}$ across the line-forming region would do it. The EW-matched EXHALE solution reaches 1.2 km s^{-1} at $20 R_p$; the paper's p-winds model, with its imposed temperature, reaches 4–7. Both are an order of magnitude short.
- *Non-thermal (turbulent) broadening.* Added in quadrature to what EXHALE already has, closing the gap takes 22.3 km s^{-1} FWHM — 9.5 km s^{-1} of turbulent velocity (σ), against the 2.9 km s^{-1} that the sound-speed convention of Sect. 4.1 supplies for helium at 5000 K. This is the one of the three that can be measured directly, and Sect. 8.2 measures it.

So the model is not short by a detail. It is short by an amount of velocity comparable to the escape speed, and no choice of composition supplies it.

8.2 The broadening the data demand, measured

The three routes above can be replaced by one measurement. Take the EW-matched EXHALE profile as solved, convolve it with a Gaussian of FWHM f in velocity, and ask which f reproduces the measured line. The convolution is applied to the excess absorption $(\max T - T) / \max T$ built from the instrument-convolved column of `tpm_He10830.txt`; depth, width and the red/blue ratio are then read off with the same three-Gaussian extractor applied to the data, and the equivalent width by the same integral over the vacuum window $10832.60\text{--}10834.20 \text{ \AA}$. So f is the only quantity varied, and every number in Table 1 is measured the way the corresponding observed number was.

Table 1: The EW-matched EXHALE profile ($\text{He}/\text{H} = 0.55$, no turbulence term) convolved with an added Gaussian of FWHM f . Observed values on the last row: red and blue depths from the C26 MCMC posterior, width and equivalent width measured here on their spectrum.

$f \text{ [km s}^{-1}\text{]}$	red [%]	blue [%]	red/blue	FWHM [$\text{\AA}$]	EW [% $\text{\AA}$]
0	4.285	0.624	6.86	0.244	1.109
10	2.393	0.321	7.45	0.435	1.109
15	1.754	0.232	7.56	0.593	1.106
20	1.366	0.180	7.61	0.762	1.090
22.3	1.237	0.162	7.62	0.841	1.076
25	1.114	0.146	7.63	0.935	1.056
observed	$1.24^{+0.22}_{-0.23}$	$0.25^{+0.14}_{-0.12}$	$6.7^{+12.7}_{-3.1}$	0.841	1.108 ± 0.030

The width is reached at $f = 22.3 \text{ km s}^{-1}$ FWHM, that is $\sigma = 9.5 \text{ km s}^{-1}$. At that one value the depth is 1.237% against the measured $1.24^{+0.22}_{-0.23}$, the blue component 0.162% against $0.25^{+0.14}_{-0.12}$, and the ratio 7.62 against $6.7^{+12.7}_{-3.1}$: depth, blue

depth and ratio all land inside the observed 1σ interval at the value fixed by the width alone. Nothing was tuned to them. Figure 4 shows the profile and the two crossings.

The number does not depend on which solution is broadened. The same crossing is at 21.9 km s^{-1} for the $\text{He}/\text{H} = 0.55$ run with the turbulence term already on, and at 22.5 km s^{-1} for solar composition — although the solar run is a quarter of the observed absorption (0.311% red, EW 0.270) and so matches the shape while missing the amount. The required broadening is therefore set by the velocity field, not by composition or by the broadening treatment already in place: the direct measurement of what Sect. 8. inferred. The equivalent width falls by 3% between $f = 0$ and $f = 22.3$ because the Gaussian wings carry absorption out of the red-pair integration window, not because absorbing power is lost.

This is descriptive, not explanatory. It states that one free parameter, applied to a solved profile, reproduces the measured line; it does not say what produces 22.3 km s^{-1} . The candidates are the ones listed below, and the measurement sharpens them by fixing the number each has to deliver.

One conversion narrows the list before it is read. If the missing 22.3 km s^{-1} is called turbulence, it has to be turbulence in gas whose sound speed the solution itself fixes. Evaluating $c_s = (\gamma kT / \mu m_H)^{1/2}$ with $\gamma = 5/3$ on the $\text{He}/\text{H} = 0.55$ output, taking μ from the solved H I/H II/He I/He II/He III and electron densities, gives $c_s = 5.5 \text{ km s}^{-1}$ at $2R_p$ ($T = 4311 \text{ K}$, $\mu = 1.94$), 3.8 at the metastable peak $5.4R_p$ (1609 K, 1.57) and 3.0 at $10R_p$ (845 K, 1.32); weighted over the line-forming region by $n(2^3S)r$, $T = 2022 \text{ K}$, $\mu = 1.59$ and $c_s = 3.9 \text{ km s}^{-1}$. Note this is the bulk-gas sound speed, the quantity a Mach number is defined against, and not the helium thermal speed that enters the line profile.

Against that, the kernel is supersonic on any convention. Read as a line-of-sight velocity dispersion, $\sigma = 9.5 \text{ km s}^{-1}$ is $\mathcal{M} = 2.4$. Read in the b -parameter convention of Lampón et al. and p-winds, where v_{turb} is the quantity added to $b^2 = 2kT/m + v_{\text{turb}}^2$, it is $v_{\text{turb}} = \sqrt{2}\sigma = 13.4 \text{ km s}^{-1}$, $\mathcal{M} = 3.4$; as an isotropic three-dimensional turbulent speed, $\sqrt{3}\sigma = 16.4 \text{ km s}^{-1}$, $\mathcal{M} = 4.2$. The $v_{\text{turb}} = c_s$ convention of Sect. 4.1 supplies only 3–4 km s^{-1} in this gas, about a third of what the data demand. That is the opposite corner of the literature reviewed there: Yan et al. (2024) argue $\mathcal{M} = 1$ is already an overestimate at a few thousand K, and Murray-Clay et al. (2009) call the energy requirement of supersonic turbulence “insurmountable”. Supplying the shortfall under the name of turbulence therefore looks physically hard to sustain, and the measurement points to the candidates of the next section — an ordered velocity field, three-dimensional flow, or heating the 1D energy equation does not carry — rather than to a missing thermal width.

8.3 The same demand, under diffusion and the adopted K_{zz}

Everything above was solved in the well-mixed limit. With binary H/He element diffusion active, helium settles out of the base unless an eddy term lifts it, and how much of it reaches the line-forming region is set by the eddy coefficient K_{zz} , which is not measured for this system. The diffusion operator is described in `binary_diffusion_design.md`; the literature range at the $1 \mu\text{bar}$ base of these runs, the ladder of solved cases, and the adoption of $K_{zz} = 1.0 \times 10^9 \text{ cm}^2 \text{ s}^{-1}$ are in `LHS1140b/kzz_decision.md`.

The composition, and how much of it K_{zz} carries

Turning the operator on moves the equivalent-width crossing of Fig. 3 away from $\text{He}/\text{H} = 0.55$, and by how much depends on K_{zz} . The composition was therefore rescanned at every decade of the eddy coefficient from $K_{zz} = 0$ to

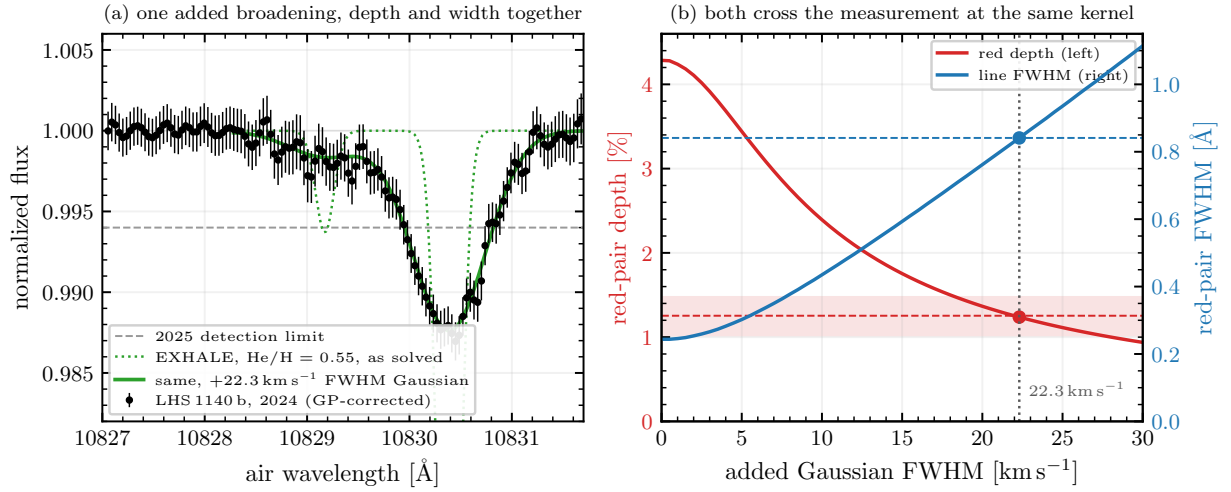


Figure 4: Left: the EW-matched EXHALE profile as solved (dotted, its core running below the plotted range) and convolved with a 22.3 km s^{-1} FWHM Gaussian (solid), against the measurement. Both are shifted by the same $v_{\text{wind}} = +2.26 \text{ km s}^{-1}$ used in the earlier figures. Right: red-pair depth (red, left axis) and red-pair FWHM (blue, right axis) as the added kernel is scanned. Dashed lines are the observed values, the shaded band the C26 1σ interval on the depth; the dotted vertical marks where the width is matched. Both curves meet the measurement at the same kernel.

$10^{11} \text{ cm}^2 \text{ s}^{-1}$, by the same route in every case: the same control input, the same direct-steady JFNK solve at $\|R\|_{\text{tol}} = 5.0 \times 10^{-3}$, the same single post-processing pass and transit synthesis, and the same red-pair equivalent width on the vacuum window $10832.60\text{--}10834.20 \text{ Å}$. Table 2 is the result; every crossing is interpolated inside a bracket of two solved runs, none extrapolated and none against a saturated line — at $K_{zz} = 10^6$, where the crossing sits highest, the equivalent width is still climbing at $\text{He}/\text{H} = 10$.

The full ladder is three regimes, not one power law (Fig. 5). Above 10^7 the degeneracy is shallow and very nearly a power law, $d \log(\text{He}/\text{H})/d \log K_{zz} = -0.136$ (-0.139 , -0.133 and -0.141 across the three decades $10^8\text{--}10^{11}$), so $\text{He}/\text{H} \propto K_{zz}^{-0.14}$. Below 10^7 the curve rolls over (-0.071 across $10^6\text{--}10^7$, -0.022 across $10^5\text{--}10^6$) onto a plateau at $\text{He}/\text{H} = 4.68$, the value the line implies when the eddy term does nothing. That plateau is where it is for a reason that is measured rather than assumed: the operator adds K_{zz} to the molecular coefficient, and these helium-rich solutions carry $D_{\text{eff}} = 1.24\text{--}1.32 \times 10^6 \text{ cm}^2 \text{ s}^{-1}$ at the base, so $K_{zz} = 10^6$ raises the base coefficient by 77%, 10^5 by 8% and 10^4 by 0.8%. At a fixed $\text{He}/\text{H} = 5$ the resulting equivalent width departs from the $K_{zz} = 0$ one by 0.005 of the measurement error at 10^4 , 0.18 at 10^5 and 1.76 at 10^6 : *no eddy coefficient at or below $10^4 \text{ cm}^2 \text{ s}^{-1}$ is distinguishable from none at all by this line*. At the other end the operator never reaches the well-mixed limit either — even at 10^{11} the inferred reservoir, 1.11, is twice the well-mixed 0.55.

Over the span the literature leaves open at the base pressure of these runs the inferred reservoir ratio moves only from 1.5 to 2.9, a factor 1.9, against a factor 9 in equivalent width across the same two decades at fixed composition: the line responds to composition much faster than it responds to K_{zz} , so a large change in the eddy coefficient is bought back by a small change in the reservoir. *The composition inferred from this line under diffusion is therefore only weakly conditional on K_{zz} , as $K_{zz}^{-0.14}$, and strongly conditional on the assumed spectrum* — the SED moves the well-mixed

Table 2: Where the red-pair equivalent width crosses the measured $1.108 \pm 0.030 \text{ \% } \text{\AA}$, at every decade of the eddy coefficient, with binary H/He diffusion active throughout. The 1σ range propagates the measurement error alone, within one K_{zz} . The decades 10^1 – 10^4 were not scanned in composition: at a fixed He/H = 5 their equivalent width differs from the $K_{zz} = 0$ one by 0.00002 to 0.005 of the measurement error, which is the entry they carry. Runs, seeds and residuals: LHS1140b/kzz_decision.md, Sect. 6.1.

$K_{zz} [\text{cm}^2 \text{s}^{-1}]$	He/H at the crossing	1σ range	bracketing runs [EW in $\text{\AA}$]
0 (molecular only)	4.68	4.54–4.83	4.6 [1.092], 4.8 [1.134]
10^1 – 10^4	not distinguishable from $K_{zz} = 0$ by this line		
10^5	4.68	4.55–4.83	4.6 [1.092], 4.8 [1.133]
10^6	4.45	4.33–4.59	4.4 [1.096], 4.7 [1.161]
10^7	3.78	3.66–3.96	3.6 [1.063], 3.8 [1.113]
10^8	2.87	2.78–2.97	2.70 [1.050], 3.00 [1.149]
10^9 (adopted)	2.09	2.03–2.16	2.00 [1.065], 2.13 [1.129]
10^{10}	1.54	1.47–1.59	1.50 [1.087], 1.60 [1.146]
10^{11}	1.11	1.07–1.15	1.00 [1.019], 1.20 [1.180]
none (well mixed)	0.550	—	Fig. 3

crossing by a factor 9, from 0.550 to 0.060 (Sect. 10.), which is five times the entire K_{zz} lever.

What K_{zz} does act on strongly is the helium profile itself, and Fig. 6 separates that from the wind. At a fixed reservoir He/H = 0.55, raising K_{zz} from 0 to 10^{10} lifts the homopause from below the base to $1.13 R_p$ and the metastable population by a factor 400, while the velocity and density profiles move by less than 30% and $\dot{M}$ by 0.23 dex. Temperature is the one bulk variable that follows, because the helium the eddy term lifts is itself a heating agent. Along the equivalent-width-matched sequence of Table 2, where each K_{zz} carries its own crossing composition, $\log_{10} \dot{M} = 7.80 \pm 0.01$ in every run: the escape rate does not carry the K_{zz} dependence at all.

The width the data demand, again

The question that remains is narrower: whether the width the data demand moves with the composition.

It does not. Repeating the measurement of Sect. 8.2 on heh2p13_diff_kzz1e9, the EW-matched diffusive solution (He/H = 2.13, red-pair EW $1.129 \text{ \% } \text{\AA}$, red depth 4.396%, FWHM $0.242 \text{ \AA}$), with the same convolution, the same three-Gaussian extractor and the same integration window, gives Table 3.

The width is reached at $f = 22.31 \text{ km s}^{-1}$ FWHM, against 22.30 km s^{-1} for the diffusion-off He/H = 0.55 solution of Sect. 8.2: the two differ by 0.01 km s^{-1} , more than three orders of magnitude below the number itself. At that kernel the depth is 1.260% against the measured $1.24^{+0.22}_{-0.23}$, the blue component 0.166% against $0.25^{+0.14}_{-0.12}$, the ratio 7.60 against $6.7^{+12.7}_{-3.1}$ and the equivalent width 1.094 against 1.108 ± 0.030 — all four inside the observed 1σ interval at the value fixed by the width alone, as in the well-mixed case. So the velocity the profile demands is unchanged by turning diffusion on and by the adopted eddy coefficient: those change how much helium is there, not how fast it moves.

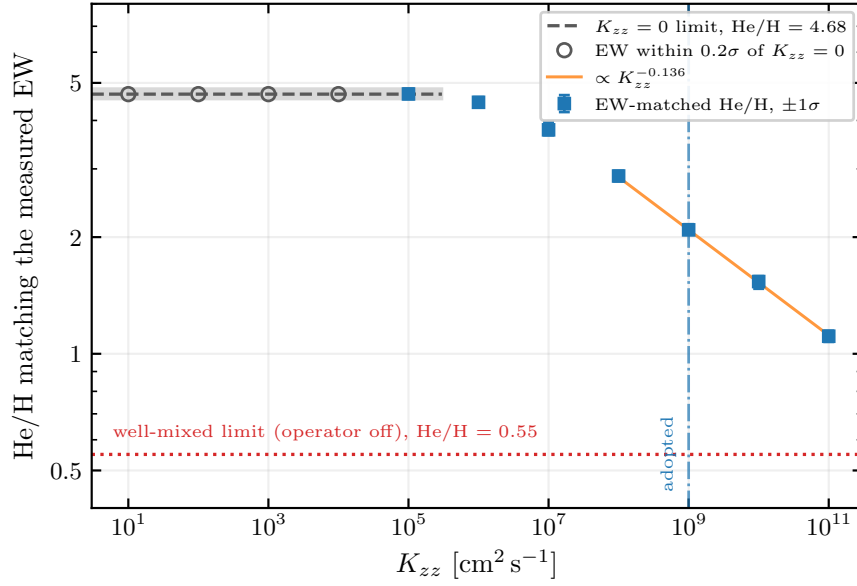


Figure 5: **The composition–mixing degeneracy in full.** Each square is the He/H reservoir ratio whose red-pair equivalent width equals the measured one, solved at one decade of K_{zz} with binary H/He diffusion active (Table 2); the error bars propagate the measurement error alone and are smaller than the symbols over most of the range. The landscape has three parts. Below $K_{zz} \simeq 10^5$ the eddy term is a small correction to the molecular coefficient the base already carries, the inference sits on a plateau at He/H = 4.68 (dashed, with its 1σ band), and the open circles mark the decades whose equivalent width differs from the $K_{zz} = 0$ one by less than a fifth of the measurement error. Between 10^6 and 10^7 the curve rolls off. From 10^8 up it is the shallow power law $K_{zz}^{-0.136}$ (orange). Neither end of the axis reaches the well-mixed crossing of 0.55 (dotted): with the operator active, helium settles at every eddy coefficient tried, and He/H inferred from this line is a statement about the assumed mixing as much as about the planet. The adopted $K_{zz} = 10^9 \text{ cm}^2 \text{ s}^{-1}$ is marked.

Table 3: The EW-matched diffusive solution (heh2p13_diff_kzz1e9: He/H = 2.13, $K_{zz} = 10^9 \text{ cm}^2 \text{ s}^{-1}$) convolved with an added Gaussian of FWHM f , measured exactly as Table 1.

f [km s ⁻¹]	red [%]	blue [%]	red/blue	FWHM [Å]	EW [% Å]
0	4.396	0.643	6.84	0.242	1.129
10	2.441	0.328	7.44	0.434	1.129
15	1.787	0.237	7.55	0.593	1.125
20	1.392	0.183	7.59	0.761	1.108
22.3	1.260	0.166	7.60	0.841	1.094
25	1.134	0.149	7.62	0.935	1.073
observed	$1.24^{+0.22}_{-0.23}$	$0.25^{+0.14}_{-0.12}$	$6.7^{+12.7}_{-3.1}$	0.841	1.108 ± 0.030

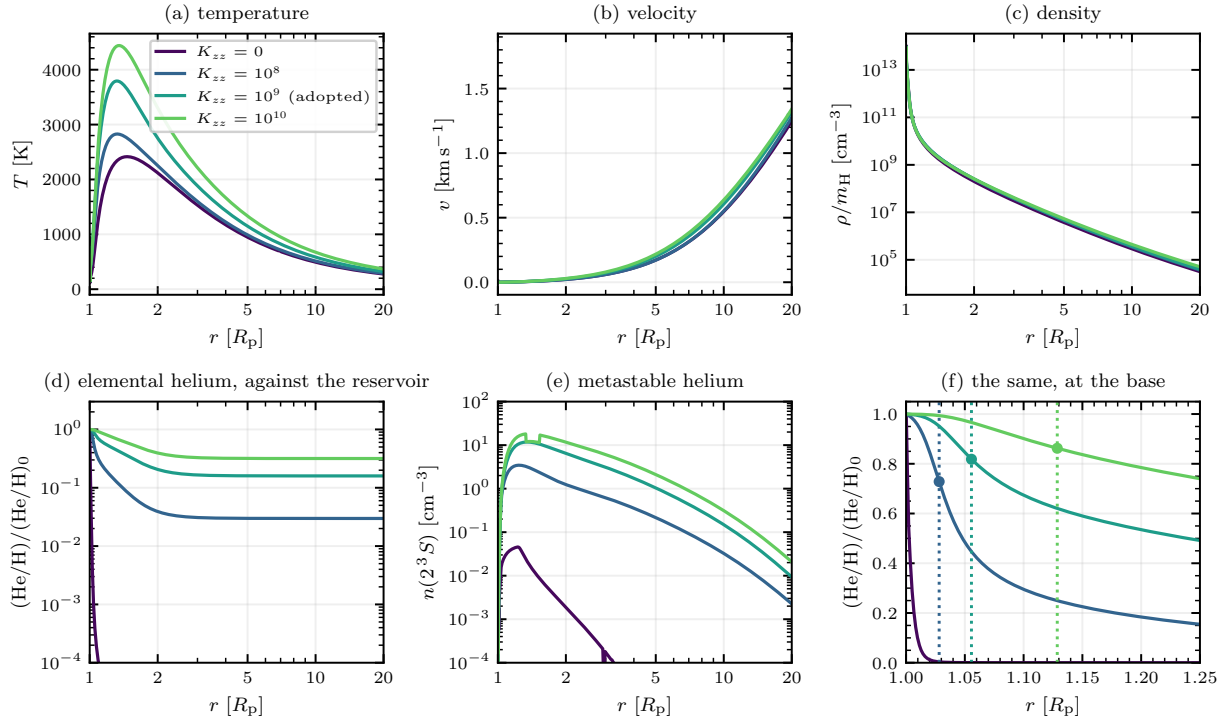


Figure 6: Four converged solutions differing only in the eddy coefficient, at a fixed reservoir $\text{He}/\text{H} = 0.55$ with binary H/He diffusion active: $K_{zz} = 0, 10^8, 10^9$ (adopted) and $10^{10} \text{ cm}^2 \text{ s}^{-1}$, dark to light. **The eddy coefficient is a composition knob, not a wind knob.** Panels (d)–(f), the elemental helium against its base reservoir value and the metastable population, span two to four orders of magnitude across the four runs — at $1.05 R_p$ the elemental ratio is 0.000, 0.485, 0.845, 0.974, and the peak $n(2^3S)$ is 0.046, 3.5, 11.8, 18.1 cm^{-3} — while in panels (b) and (c) the velocity at $10 R_p$ moves by 16% and the density at $2 R_p$ by 28%, and $\log_{10} \dot{M}$ by 0.23 dex. Panel (a) is the exception: T at $1.5 R_p$ rises from 2412 to 4276 K, because the helium the eddy term lifts is itself absorbing. Panel (f) repeats (d) on a linear axis over the base, where the dots and dotted lines mark each run’s homopause — the radius at which its own molecular coefficient equals its K_{zz} — at 1.028, 1.056 and 1.129 R_p , with the $K_{zz} = 0$ homopause below the base.

Figure 7(a) displays the radial number density profiles of the hydrogen and helium species in this diffusive steady state, and Figure 7(b) shows the resulting broadened transit spectrum overlay.

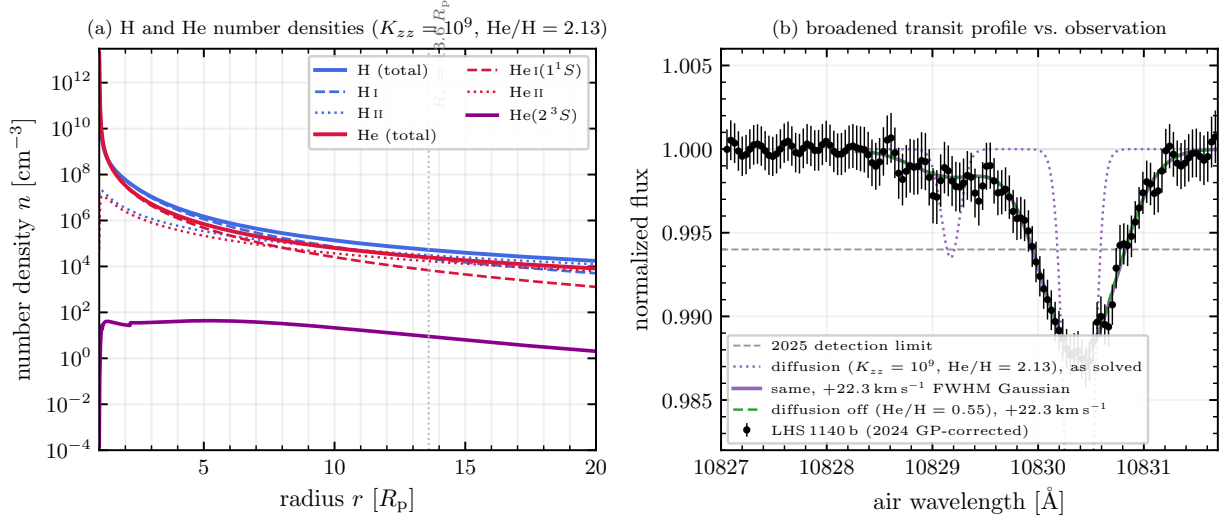


Figure 7: **The EW-matched diffusive solution** (heh2p13_diff_kzz1e9: $\text{He}/\text{H} = 2.13$, $K_{zz} = 10^9 \text{ cm}^2 \text{ s}^{-1}$). **(a)** Absolute number densities of hydrogen species (total H, H I, H II) and helium species (total He, He I(1^1S), He II, and metastable He(2^3S)) as a function of planetary radius r [R_p]. At the lower boundary ($r = 1.0 R_p$), $\text{He}/\text{H} = 2.13$; binary element diffusion causes helium to settle outward relative to hydrogen, with the local abundance falling to $\text{He}/\text{H} \approx 0.57$ at $2 R_p$ and 0.47 at $10 R_p$. Dotted vertical line marks the stellar disk radius ($13.6 R_p$). **(b)** The transmission spectrum as solved (dotted purple, its core running below the plotted range) and convolved with the 22.3 km s^{-1} FWHM Gaussian that reproduces the measured width (solid purple), against the measurement; the diffusion-off $\text{He}/\text{H} = 0.55$ curve at its own matched kernel (dashed green) is repeated from Fig. 4. The two broadened curves are indistinguishable — their kernels differ by 0.01 km s^{-1} . All model curves are shifted by the fitted $v_{\text{wind}} = +2.26 \text{ km s}^{-1}$.

8.4 Broadening cannot repair a deficit in equivalent width

The same scan was applied to the p-winds curves, in their own vacuum frame and with the same three-Gaussian extractor and the same integration window (Table 4). The convolution conserves absorbing power: it redistributes the excess in velocity and leaves the equivalent width unchanged, apart from the few percent that the wings carry out of the integration window. Whether one kernel can reproduce depth and width together is therefore decided before the scan starts, by the equivalent width the curve already has at $f = 0$.

The p-winds best fit carries 0.837% $\text{\AA}$ against the measured 1.108 ± 0.030 , 24% short. At the kernel that reproduces the width, $f = 17.8 \text{ km s}^{-1}$, the red depth is 0.926% against $1.24^{+0.22}_{-0.23}$, about 1.4σ low. What the broadening does is move the disagreement rather than remove it: unbroadened, the curve is too deep in the blue component (0.534% against $0.25^{+0.14}_{-0.12}$, some 2σ high) and too narrow; broadened to the measured width, the blue component falls to 0.339% , inside 1σ , and the red component drops below the measurement. The blue side is genuinely improved; the red side is not, and the total is fixed.

Table 4: The same added-Gaussian scan applied to the p-winds curves, with the EXHALE row of Table 1 repeated for comparison. The kernel is the one that reproduces the measured width, $0.841 \text{ \AA}$; the depths and the ratio are read at that kernel. Equivalent widths are for the unbroadened curve, over the vacuum window $10832.60\text{--}10834.20 \text{ \AA}$.

curve	EW($f=0$) [% $\text{\AA}$]	f [km s^{-1}]	red [%]	blue [%]	red/blue
p-winds best fit, turbulence on	0.837	17.8	0.926	0.339	2.73
p-winds best fit, turbulence off	0.701	20.1	0.777	0.301	2.58
p-winds depth-matched, 6100 K	0.480	20.2	0.535	0.140	3.82
EXHALE, He/H = 0.55	1.109	22.3	1.237	0.162	7.62
observed	1.108 ± 0.030	—	$1.24^{+0.22}_{-0.23}$	$0.25^{+0.14}_{-0.12}$	$6.7^{+12.7}_{-3.1}$

The intensity ratio barely moves, 2.57 at $f = 0$ against 2.73 at the matched kernel. It is the amplitude ratio of a saturated pair, set by how the two components share the optical depth, and a common convolution acts on both alike. It stays well below the measured $6.7^{+12.7}_{-3.1}$, though the observed interval is wide enough that this alone is not a discriminator.

Matching the depth as well would require 24% more absorbing power, and that is not something a kernel supplies: it is a change in $\dot{M}$, temperature or composition. Fitting the broadening as a free parameter alongside those — which is what the measurement asks for — would presumably shift the C26 parameter estimates, since the composition is constrained through the same absorbing power that would have to rise.

The asymmetry with EXHALE should not be read as one code outperforming the other. The EXHALE curve entered the scan with its equivalent width already set to the observed value by the choice of He/H = 0.55 (Sect. 8.2), so only the shape remained free, and one kernel settled it. The p-winds curve entered at the composition C26 report, whose equivalent width falls short, so no kernel could settle both. The two scans test the same physical statement — broadening redistributes, it does not create — from two different starting points.

8.5 The adiabatic-cooling population bump, looked for

Schulik & Owen (2025) propose that the missing width is already contained in the metastable population itself. Writing the steady state of their Eq. (1) as

$$n_{2^3S}(T_e) = \frac{\alpha(T_e)}{q(T_e) + t_{\text{ion}}/n_e} n_{\text{He}^+}, \quad (4)$$

they note that as adiabatic expansion drops the temperature “from $\simeq 10^4 \text{ K}$ to $\simeq 10^3 \text{ K}$ ”, “a change in slope in the population occurs as the dominant triplet loss rate $q(T_e)$ drops significantly at lower temperatures”—the metastable fraction climbs above the curve computed with the rates frozen at the peak temperature. The rise is “quenched” where the advection timescale overtakes the excitation to 2^1S , after which $\dot{m}_{2^3S} = 4\pi r^2 v n_{2^3S} m_{\text{He}}$ falls through ionization and the $2^3S \rightarrow 1^1S$ decay. They credit Yan et al. (2022) and Biassoni et al. (2024) with first identifying the effect, and their conclusion states the condition it must satisfy to be observable: “The helium triplet population bump must be situated at high velocities to help explain anomalous line widths.”

EXHALE carries the temperature dependence the mechanism needs, and carries it in exactly the balance above. The

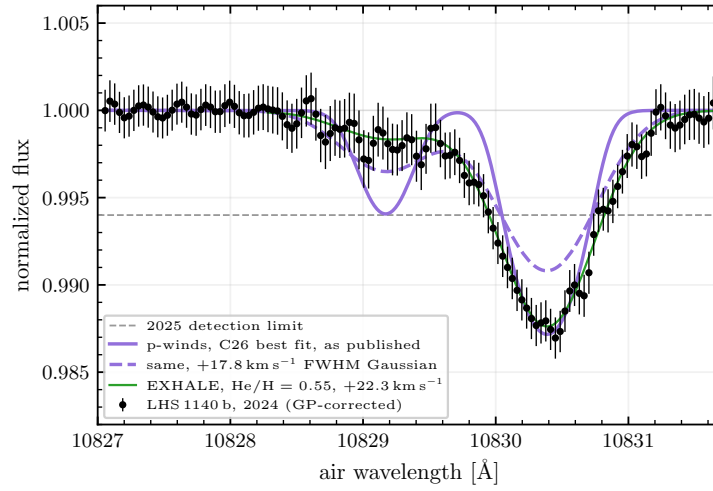


Figure 8: The C26 p-winds best fit as published (solid purple) and convolved with the 17.8 km s^{-1} FWHM Gaussian that reproduces the measured width (dashed purple), against the measurement; the EXHALE $\text{He}/\text{H} = 0.55$ curve at its own matched kernel (22.3 km s^{-1} , thin green) is repeated from Fig. 4. Broadened to the observed width, the p-winds curve passes above the red core: the width is matched and the depth is not. All curves are shifted by the same $v_{\text{wind}} = +2.26 \text{ km s}^{-1}$.

metastable row (`tr_triplet_row`, `ion_residual_core.f90:72`) is

$$n_e (n_{\text{He}^+} \alpha_{2^3S} + n_{\text{HeI}(1^1S)} q_{13}) = n_{2^3S} [\Gamma_{2^3S} + A_{31} + n_{\text{HI}} Q_{31} + n_e (q_{31a} + q_{31b} + b_{2^3S})], \quad (5)$$

with every coefficient a function of T : $\alpha_{2^3S} = 2.10 \times 10^{-13} (T/10^4)^{-0.778}$ (`Cool_coeff.f90:970`), the collisional losses q_{31a} ($2^3S \rightarrow 2^1S$, 0.80 eV) and q_{31b} ($2^3S \rightarrow 2^1P$, 1.40 eV) as Bray (2000) collision strengths folded with $\exp(-\Delta E/kT)$ (`Cool_coeff.f90:905` and `928`), the Penning term Q_{31} as the Taylor et al. (2025) two-branch fit (`Cool_coeff.f90:988`), collisional ionization of the metastable (`Cool_coeff.f90:1030`), and the radiative decay $A_{31} = 1.272 \times 10^{-4} \text{ s}^{-1}$ (`util_ion_eq.f90:1192`). Between 5000 and 1000 K the loss $q_{31a} + q_{31b}$ falls by a factor 1.3×10^3 while α_{2^3S} rises by a factor 3.5 , so the ingredients of the bump are present and are not an imposed parameterization.

`LHS1140b/bump_analysis.py` measures the result. It reconstructs the balance above in Python from the solved profiles—reproducing the solved n_{2^3S} to a relative 3×10^{-16} , which confirms the rate transcription and lets the one unresolved term, the 2^3S photoionization rate Γ_{2^3S} , be recovered from the solution itself—and then recomputes the reference curve with α , q_{13} , q_{31a} , q_{31b} and Q_{31} frozen at the peak temperature.

The bump is there. In every case the metastable fraction $f_3 = n_{2^3S}/n_{\text{He}}$ rises monotonically outward across the whole cooling leg, from the temperature maximum to a maximum of f_3 at $1.1\text{--}16.2 R_p$ —a rise of $1.4\text{--}4.2 \times 10^3$ in the equilibrium profiles and $22\text{--}236$ in the advection-corrected ones—and the rise is attributable to the temperature dependence in the way Schulik & Owen describe: against the same balance with the rates frozen at the peak temperature, the solved population stands a factor $1.6\text{--}31$ high at the density maximum and up to $8.8\text{--}156$ high further out. On the GJ 1132 proxy at the equivalent-width-matched composition the maximum sits at $5.42 R_p$ where $T = 1608 \text{ K}$, against

Table 5: The metastable bump in the converged solutions. Columns 2–4 locate the n_{2^3S} maximum in the equilibrium profile and give the temperature and outflow speed there; the last three characterize the absorber-weighted ($w = n_{2^3S} r^2 dr$) distribution of $|v_r|$ over the advection-corrected 2^3S column that the transit tool integrates. The He/H = 1000 solutions have no outward density maximum—their n_{2^3S} peaks at the base—and are marked †. Those two rows are corrected values, recomputed on the advection-corrected profiles that the gate fix of Update_EXHALE.md Sect. 72 produced; the He/H ≤ 1 rows are unaffected by it.

case	r_{peak} [R_p]	T [K]	v [km s ^{−1}]	med. $ v_r $ [km s ^{−1}]	> 2 [%]	> 5 [%]
<i>GJ 1132 proxy SED</i>						
He/H = 0.55	5.42	1608	0.27	0.38	0.0	0.0
solar He/H	3.02	1876	0.07	0.41	0.0	0.0
He/H = 1000	1.05 [†]	2550	0.00	0.00	0.0	0.0
<i>GJ 699 proxy SED</i>						
He/H = 0.06	5.42	1015	1.40	2.53	64.9	0.0
solar He/H	5.50	1047	1.41	2.51	64.3	0.0
He/H = 1000	1.05 [†]	2529	0.00	0.00	0.4	0.0

$T_{\text{max}} = 5043$ K at $1.45 R_p$: the temperature has fallen to 0.32 of its peak, which is the regime the mechanism asks for. The quench is visible too—in the advection-corrected profiles the density maximum has moved inward to 1.05 – $1.64 R_p$ and n_{2^3S} then falls monotonically, so the outflow appears to be sweeping the metastable out at least as fast as recombination resupplies it, and $\dot{m}_{2^3S}$ drops past the peak by 31% at the GJ 1132 equivalent-width-matched composition and by 98–99.9% on the helium-dominated He/H = 1000 runs, while the remaining cases carry it outward without loss.

The velocity condition, however, is not met, and not narrowly. At the bump the outflow is moving at 0.27 km s^{−1} on the GJ 1132 proxy and 1.4 km s^{−1} on the GJ 699 proxy. Weighting by absorbers over the whole 2^3S column, the median $|v_r|$ is at most 0.4 km s^{−1} and 2.5 km s^{−1} on the two spectra, and the largest $|v_r|$ anywhere in the column is 1.6 and 3.9 km s^{−1}. No part of the column on either spectrum reaches 5 km s^{−1}, against the $\sigma = 9.5$ km s^{−1} of extra broadening the measurement demands (Sect. 8.2). Radial velocity is the generous choice here: the line-of-sight projection of a radial field over the transit chord is smaller still. Figure 9 shows the three statements together.

So the bump exists in these solutions and contributes nothing measurable to the width. That is consistent with what Schulik & Owen themselves report for the coolest hosts: for their M2 case they find that “the transit signal probes the same velocity region for both treatments of thermodynamics, making the line insensitive to velocity”, and they conclude that K stars, not M stars, are the favorable testbeds—G-type hosts suppress the bump by photoionizing the triplet, and M-type hosts have weaker tidal fields and so rarely drive fast enough outflows. LHS 1140 is an M4.5 dwarf, cooler than the M2 they model, and its planet is a temperate super-Earth rather than a hot giant. The mechanism appears to be present and correctly captured; the host is the wrong one for it to help.

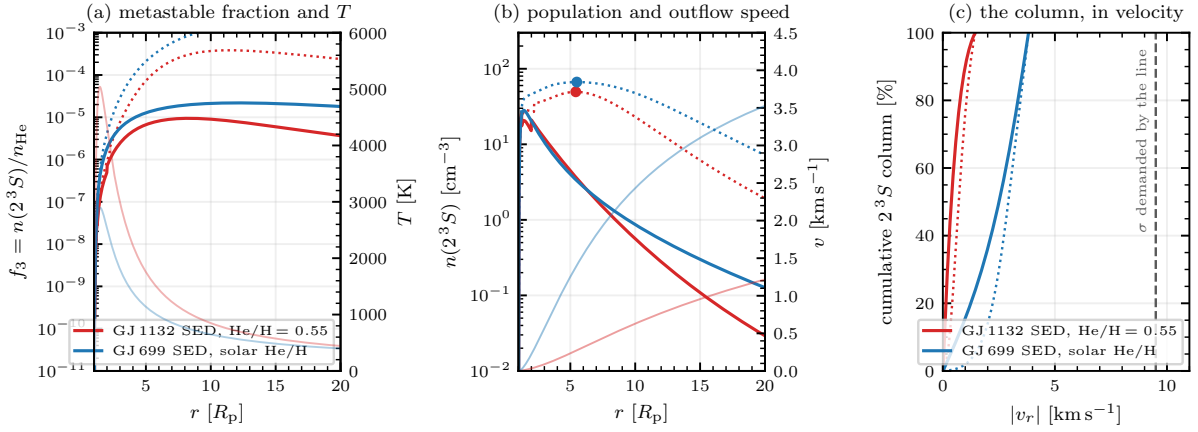


Figure 9: The adiabatic-cooling metastable bump in the EXHALE solutions (solid: advection-corrected, the profiles the transit integral uses; dotted: local equilibrium; faint curves on the right-hand axes). (a) The metastable fraction rises outward over the whole region where T falls, which is the bump in the sense of Schulik & Owen (2025). (b) The equilibrium n_{2^3S} peaks at $5.4\text{--}5.5 R_p$ (markers); advection moves the maximum inward and quenches the outer rise. (c) The absorber-weighted distribution of $|v_r|$ over the 2^3S column: the whole column lies below 4 km s^{-1} , far short of the $\sigma = 9.5 \text{ km s}^{-1}$ the measured width demands.

8.6 This is a published shortfall

Taylor et al. (2026), with a model of the same class, reproduce the HD 189733 b line core but need $\sim 12 \text{ km s}^{-1}$ of imposed non-thermal broadening for the width. Rumenskikh et al. (2022), in 3D, instead imposed reduced hydrogen cooling and $10\text{--}20 \text{ km s}^{-1}$ zonal jets. Schulik & Owen (2025) argue the width can come from a metastable population bump produced by adiabatic cooling, with no free parameter. The present result reproduces that pattern on a different, much cooler planet: a self-consistent energy balance produces a line far too narrow, and the literature’s responses are all ways of adding velocity that the 1D energy equation does not generate.

8.7 What would have to change

The measured line wants gas that is fast enough to be broad but not so extended that it absorbs too much. EXHALE at this XUV flux delivers neither. The candidates, in the order they should be tested:

1. *More energy* — tested (Sect. 10.), and it does not do it. The GJ 699 proxy carries five times the XUV of the GJ 1132 proxy used here, and the paper itself runs both. Solving the scan again on it raises $\dot{M}$ sevenfold and moves the composition estimate by an order of magnitude, but leaves the required extra broadening at 22.08 km s^{-1} against 22.30 : the width gap is not an energy-input problem.
2. *A different helium distribution* — tested (Sect. 8.3), and it does not do it either. Letting helium settle against binary diffusion and lifting it back with an eddy term redistributes the absorber by orders of magnitude and moves the inferred composition with it, but the required broadening stays at 22.31 km s^{-1} : it changes how much helium is there, not how fast it moves.

3. *Heating the code does not carry.* Taylor et al.’s trapped-ion turbulence, or any magnetically mediated dissipation, is outside a 1D hydrodynamic energy equation by construction.
4. *Geometry.* A line-of-sight velocity field from day–night or zonal flow is a 3D effect; Rumenskikh et al. reach the observed width with 10–20 km s^{−1} jets, which is the right order.
5. *Adiabatic-cooling population bump.* Looked for (Sect. 8.5): the bump is present in these solutions, but it sits at 0.3–1.4 km s^{−1}, so it does not supply width.

The paper’s composition constraint should be read in the same light: it inherits the isothermal assumption. Impose a 6100 K structure and only helium-rich gas absorbs enough; solve the energy balance on this spectrum and, in the well-mixed limit, the same absorption is reached near He/H $\sim$ 0.5 — and near 0.06 on the other proxy spectrum, or near 2 once helium is allowed to settle against the adopted eddy coefficient. No answer on that list is secure while the width is unexplained, because the width is what tells us whether the line-forming gas is the gas either model thinks it is.

9. The best each model can do

Figure 10 puts only the two best models on the measurement: the paper’s p-winds fit at its own best fit (the MCMC medians), and the EXHALE solution whose equivalent width matches. They fail in opposite directions. p-winds carries roughly the right width because its temperature is imposed where the line forms, but overshoots the blue component by a factor of about three. EXHALE carries the right total absorption but concentrates it into a line a third as wide as observed, so its core is too deep. Only the second failure can be undone by broadening alone (Sects. 8.2 and 8.4).

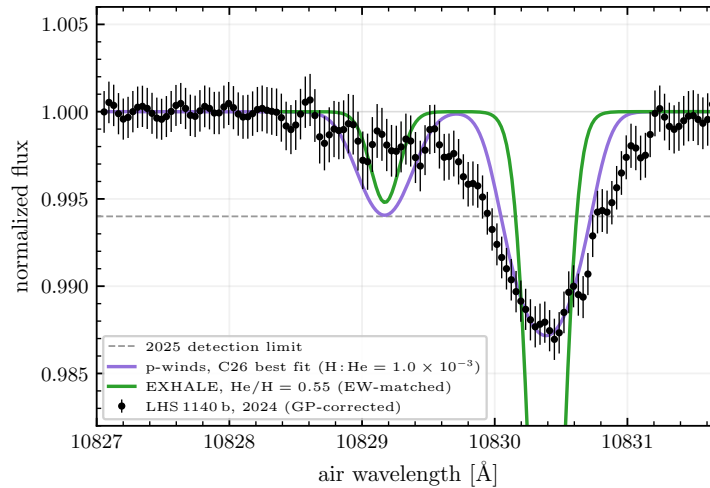


Figure 10: The two best models over the 2024 spectrum, in the presentation of the paper’s Fig. 4: p-winds at the C26 best fit (the MCMC medians) with the authors’ settings, and EXHALE at the equivalent-width-matched composition. Both are shifted by the fitted $v_{\text{wind}} = +2.26$ km s^{−1}.

Figure 11 shows what that EXHALE solution actually is. It is not a fast, hot outflow: the temperature peaks at 5043 K

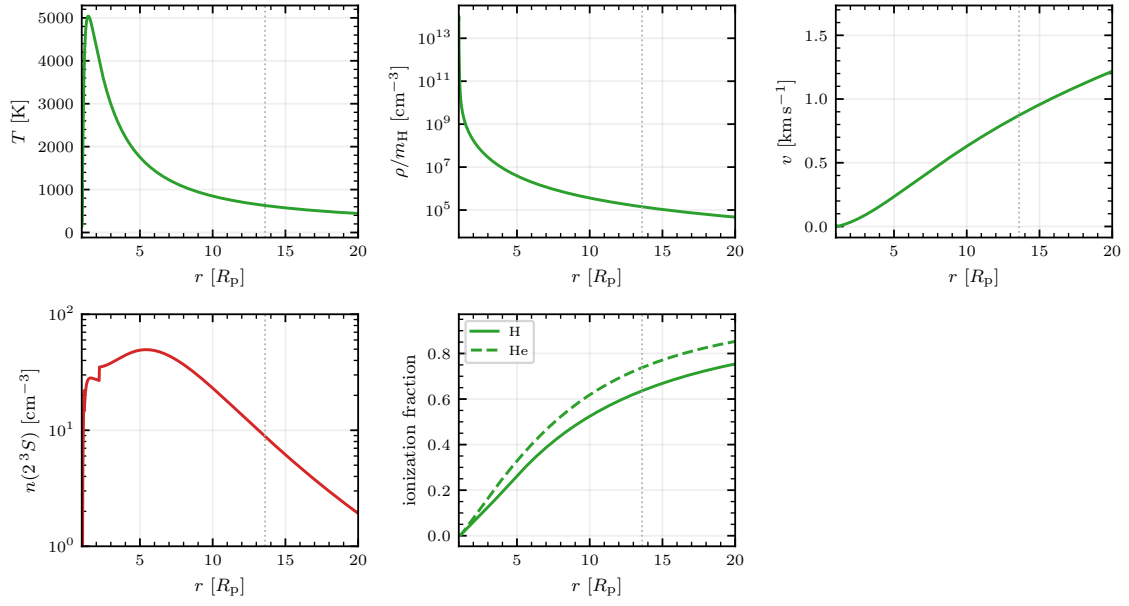


Figure 11: The EW-matched EXHALE solution ($\text{He}/\text{H} = 0.55$): temperature, density, velocity, metastable helium density, and ionization fractions. Dotted vertical line: the stellar radius.

near $1.45 R_p$ — comparable to the isothermal value p-winds imposes — but then falls by adiabatic expansion to below 1000 K over the region that forms the line, and the velocity reaches only 1.2 km s^{-1} at $20 R_p$. Hydrogen is fully ionized above $\sim 3 R_p$ while helium stays largely neutral, and the metastable population peaks well out at $5.4 R_p$, $n(2^3S) \simeq 50 \text{ cm}^{-3}$. That combination — a large, cold, slow, metastable-rich envelope — is precisely what produces the right equivalent width with the wrong profile: plenty of absorbers spread over most of the stellar disk, but almost no velocity spread to distribute the absorption in wavelength.

10. The other proxy: five times the XUV

The first item on the list above — more energy — can be tested without adding any physics, because the paper itself runs two proxy spectra. Both stand in for LHS 1140’s unobserved XUV: GJ 1132 (used everywhere above) and GJ 699, scaled to the orbit by the catalog X-ray flux ratios $A_{1132} = 0.59$ and $A_{699} = 0.056$ following the authors’ script, the same convention adopted in Sect. 2.3. Integrated over $10\text{--}1300 \text{ \AA}$ at the orbit of the planet, the GJ 699 file carries $169.9 \text{ erg s}^{-1} \text{ cm}^{-2}$ against 33.5 for the GJ 1132 one: a factor 5.1 in XUV input.

On the EXHALE side nothing else changes. Each GJ 699 case is the matching GJ 1132 input with one key replaced, `Spectrum file::`; the luminosity keys in the file are irrelevant because `sed_read` overwrites them with the integral of the spectrum it reads. Each case is started from the converged solution of the same composition on the GJ 1132 spectrum and finished with JFNK, and all seven reach `info = 0` with $\|R\| < 10^{-3}$. One case needed a different seed: $\text{He}/\text{H} = 0.55$ stalled when started from its GJ 1132 counterpart and converged when restarted from the already-solved $\text{He}/\text{H} = 0.25$ GJ 699 solution.

10.1 Escape rate: the retrieval value, on this spectrum

The brighter spectrum moves $\dot{M}$ where the composition never did:

	GJ 1132 proxy	GJ 699 proxy
F_{XUV} (10–1300 Å) [erg s ⁻¹ cm ⁻²]	33.5	169.9
$\log_{10} \dot{M}$ [g s ⁻¹], He/H ≤ 1	7.75–7.80	8.58–8.62
$\dot{M}$ [g s ⁻¹]	6×10^7	4.0×10^8
C26 retrieval on the same proxy	2.03×10^8	$4.22^{+1.14}_{-1.00} \times 10^8$

A factor 5.1 in XUV buys a factor $\simeq 7$ in $\dot{M}$ — steeper than linear, as expected when a larger fraction of the input goes into the wind rather than into radiation. The result is that on this spectrum EXHALE’s escape rate agrees with the paper’s retrieved value, $\log_{10} \dot{M} = 8.59$ against 8.63, inside 0.02–0.04 dex and far inside the retrieval’s own ± 0.11 dex. It is also 1.4 times the energy-limited estimate the paper quotes for this proxy, 2.90×10^8 g s⁻¹. On the GJ 1132 spectrum the same code fell a factor 3 short of the retrieval. Whether a solved wind and an imposed one agree on $\dot{M}$ therefore depends on which proxy is assumed, and the disagreement reported in Sect. 8. is not a fixed property of the code.

10.2 Structure and line

Seven compositions were solved on this spectrum, two of them (He/H = 0.04 and 0.06) placed specifically to bracket the equivalent-width crossing:

He/H	$\log_{10} \dot{M}$ [g s ⁻¹]	T_{max} [K] (at r/R_p)	$v(20R_p)$ [km s ⁻¹]	$n(2^3S)_{\text{max}}$ [cm ⁻³] (at r/R_p)	red [%]	FWHM [Å]	EW [% Å]
0.04	8.58	2605 (1.49)	3.61	30.9 (5.4)	2.65	0.270	0.761
0.06	8.59	2748 (1.48)	3.56	47.2 (5.4)	3.83	0.270	1.104
0.083	8.61	2896 (1.48)	3.53	66.7 (5.5)	5.21	0.272	1.507
0.25	8.62	3861 (1.44)	3.21	213 (5.9)	11.98	0.279	3.547
0.55	8.60	5079 (1.39)	2.83	441 (6.6)	16.64	0.288	5.092
1	8.59	6094 (1.38)	2.58	616 (7.2)	17.28	0.298	5.479
10 ³	8.43	9351 (1.38)	2.07	1.4×10^4 (1.0)	9.53	0.342	3.479
observed	—	—	—	—	1.254	0.841	1.108 ± 0.030

The wind is hotter and faster than on the fainter spectrum — at He/H = 0.06 it reaches 3.6 km s⁻¹ at $20R_p$ against 1.2 for the equivalent-width-matched GJ 1132 solution — and the metastable population still peaks far out, near $5.4R_p$. As in Sect. 5., the He/H = 10³ row is a corrected value (red 105.1% → 9.53%, EW 35.87 → 3.479 % Å after the advection-gate fix, Update_EXHALE.md Sect. 72); the six helium-poorer rows, which are the ones the crossing is interpolated in, are unaffected.

10.3 The composition estimate moves by an order of magnitude

Interpolating the equivalent width in log–log across the seven solutions and solving for the measured value gives

$$\text{He}/\text{H} = 0.060 \quad (0.059\text{--}0.062 \text{ on the } EW \text{ } 1\sigma), \quad \text{H}:\text{He} = 16.6, \quad (6)$$

with the turbulence term on, 0.060 — again a shift below the percent level, so the crossing is set by the absorbing column and not by the broadening treatment. The bracketing runs are what fix this: the two compositions straddle the measured band directly ($EW = 0.761$ and 1.104 against $1.108 \pm 0.030 \text{ } \%$ Å), so the crossing is measured rather than extrapolated from the solar case.

This is a shift of one order of magnitude from the $\text{He}/\text{H} = 0.55$ the same code gives on the GJ 1132 proxy, and it crosses the solar value on the way: the brighter spectrum wants a helium-*poor* atmosphere, $0.72\times$ solar, where the fainter one wanted a helium-rich one, $6.6\times$ solar. The measurement cannot arbitrate, because the two proxies are two guesses at the same unobserved spectrum and both reproduce the observed equivalent width once composition is free. The conclusion to draw is therefore about the inference, not about the planet: what the helium line constrains is the product of composition and XUV input, and any composition quoted from it — the paper’s $\text{H}:\text{He} \lesssim 10^{-3}$ as much as the values here — is conditional on the assumed SED. On the present evidence the composition inference is SED-limited.

10.4 The width does not move

The width is the one thing the extra energy does not fix. Across the whole GJ 699 scan the red-pair FWHM is $0.270\text{--}0.342 \text{ } \text{\AA}$ ($0.304\text{--}0.427$ with the turbulence term), against the measured $0.841 \text{ } \text{\AA}$. Measuring the deficit the way Sect. 8.2 does — convolving with a Gaussian and asking which width reproduces the observation — the composition at the crossing needs 22.08 km s^{-1} FWHM of added velocity spread, against 22.30 km s^{-1} for the equivalent-width-matched GJ 1132 solution and 22.31 for the diffusive one of Sect. 8.3. A factor 5.1 in XUV input, a factor 7 in $\dot{M}$ and a factor 3 in outflow velocity at $20R_p$ change the required broadening by 1%.

The reason is visible in the table: the outflow velocity that grows is radial and small compared with the missing spread, while the line-forming region stays cold. The candidate “more energy” is therefore tested and rejected as an explanation of the width. It moves $\dot{M}$ and it moves the composition estimate; it does not move the profile. The remaining candidates — heating outside a 1D hydrodynamic energy equation, 3D velocity fields, and an adiabatic-cooling metastable bump — are unaffected by this test; the last of them has since been looked for on both spectra (Sect. 8.5) and does not help either.

11. Closing the loop: the composition as an output

Every number above took the helium abundance as an input. The scans of Sects. 5. and 8.3 chose it, solved the wind, and asked what the line did; the reservoir at the base was a parameter set by hand, and the lower atmosphere below the $1 \mu\text{bar}$ matching level was a handful of scalars. This section removes that freedom on one axis: the lower atmosphere is solved as a photochemical column, handed over as a profile, and the elemental fluxes are iterated until they are continuous across the matching level. The composition at the match is then a solution of the coupled problem rather than

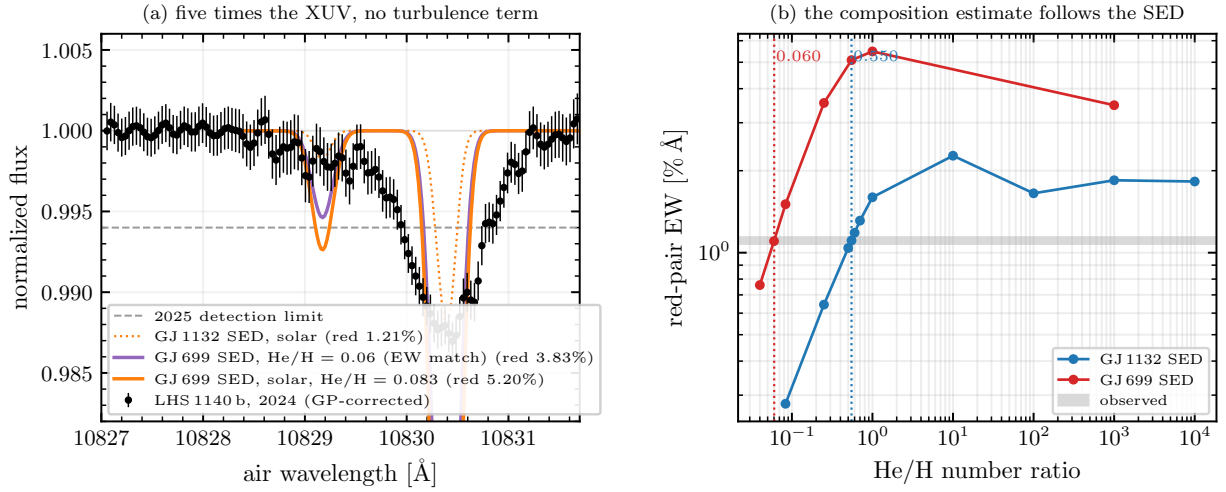


Figure 12: The GJ 699 proxy, five times the XUV of the GJ 1132 proxy used elsewhere in this memo. Left: the measurement with the EXHALE solution nearest the equivalent-width crossing ($\text{He}/\text{H} = 0.06$) and the solar case on this spectrum, with the GJ 1132 solar case dotted for contrast; all curves without the turbulence term and shifted by the fitted $v_{\text{wind}} = +2.26 \text{ km s}^{-1}$. The lines are deeper and no wider. Right: red-pair equivalent width against composition for the two spectra with the measured band and the two crossings; the composition the line implies moves from 0.550 to 0.060 when the assumed SED is changed.

a number chosen for it. The implementation and its tests are recorded in docs/phase_e_flux_closure_design.md and docs/Update_EXHALE.md, Sects. 76–79; the runs are LHS1140b/exhale/flux_closure/.

11.1 The setup

The column below the match is solved with Photochem (Wogan 2025) on the Zahnle reaction set restricted to H/He/N/O/C. The temperature is not prescribed: clima’s AdiabaticClimate is run with solve_for_T_trop on, so the stratospheric temperature is the skin temperature of the solution, and the background gas is chosen by abundance rather than by name — on this atmosphere that is helium, not H_2 . The climate solve is taken from a deep boundary at 20 bar, which is the shallowest one at which the photochemical model bottom fits inside the climate column. Chemistry and climate are run on the same elemental vector, He/H starting at 2.09 — the $K_{\text{zz}} = 10^9$ crossing of Table 2 — with Lodders C, N and O, on the GJ 1132 proxy spectrum at the planet’s orbit and at the adopted constant $K_{\text{zz}} = 10^9 \text{ cm}^2 \text{ s}^{-1}$. The result is handed to EXHALE as a profile of 102 levels from 16.73 bar to 1.10×10^{-8} bar, carrying p , r , T , n , ρ , K_{zz} , the molecular fractions and the elemental reservoirs He/H, C/H, N/H and O/H, matched to the wind at $1 \mu\text{bar}$; Sect. 11.2 reads that solution column by column, and says which of the columns the wind is actually built from. The loop is then closed in the other direction: the elemental escape fluxes EXHALE computes are imposed back on the photochemical model at its top — $\phi_{\text{El}} = \Phi_{\text{El}}/(m_{\text{El}} 4\pi r_{\text{top}}^2)$, helium on He and hydrogen on H_2 at $\phi_{\text{H}}/2$, the dominant carriers, the atomic-H share at that level being 10^{-4} — and the pair is iterated by damped Picard ($\omega = 0.5$) on the residual $|F_{\text{measured}} - \Phi_{\text{trial}}|/|F_{\text{measured}}|$ to a tolerance of 5%.

Two things about that measurement are worth stating, because they decide whether it means anything. The elemental

flux is read in the *steady* window $r \geq r_{\text{esc}}$, not in the interval the two models share: at a steady state $4\pi r^2(\rho X v + J)$ does not depend on radius, so a flux measured there is the flux through the matching level. The overlap window cannot carry it — from the match to the profile top the column spans 1.96 decades of pressure in $0.0098 R_p$, entirely inside the region where the wind's mass flux is not yet its own, and the flux across it has a radial spread of 13 in F_{H} and 35 in F_{He} . In the steady window the same solution holds 191 faces at spreads of 0.44% and 0.46%. And that flatness is a statement about how far the wind was converged, which was measured rather than assumed: at $\|R\|_{\text{tol}} = 10^{-3}$ the same solution spreads by 7.4%, which would fail the closure's own validity test, and continuing it to 10^{-4} brings it to 0.50% and moves $\log_{10} \dot{M}$ from 7.47 to 7.50. The closure runs the wind at 10^{-4} .

Not verified: the cross-check with a second chemistry code. The same handoff was set up with VULCAN (Tsai et al. 2017) on the production profile's own $T(p)$ and K_{zz} , and VULCAN did not converge on this cold helium-rich column: its convergence measure rose from 2.65 to 2.75 against a criterion of 0.01 with the timestep pinned, the limiting cell being H_2O at the cold trap, and the run exited on its step limit. No VULCAN composition at the match was produced and no wind was run on one, so the cross-code spread is not measured here. The record is LHS1140b/lower_profile_vulcan/vulcan_work/STOPPED.txt.

11.2 The column the chemistry returns

The settings above produce one file, and what EXHALE does on the profile branch rests entirely on it, so this section reads the solution itself. Figs. 13 and 14 draw every column of the schema except the two elemental-flux columns, which are constants carrying the trial fluxes the header states. The file measured here is LHS1140b/lower_profile/lower_atmosphere_profile.dat, the one-way handoff at iteration 0; the closed arms of Sect. 11.6 carry their own copies and are compared at the end of this section.

The grid and the thermal structure. 102 levels from 16.73 bar to 1.098×10^{-8} bar — the photochemical model's own domain, sitting inside the 20 bar column the climate step was taken from. Temperature runs from 520.3 K at the deep boundary to a floor of 185.05 K, which it reaches at 0.94 bar and holds at every level above: the stratosphere is isothermal at the skin temperature, and the tropopause the climate solve reports, 1.031 bar, is one level below the first isothermal level. K_{zz} is $10^9 \text{ cm}^2 \text{ s}^{-1}$ at all 102 levels, so the eddy coefficient the wind runs at and the one the column was mixed at are the same constant. The column spans 553.6 km in radius, 0.15608 to $0.16400 R_J$, with the $1 \mu\text{bar}$ match at $0.16240 R_J = 1.782 R_{\oplus}$ — which is what fixes the base radius of the wind. Density falls from $n = 2.33 \times 10^{20} \text{ cm}^{-3}$ ($\rho = 1.40 \times 10^{-3} \text{ g cm}^{-3}$) to $4.30 \times 10^{11} \text{ cm}^{-3}$ (2.58×10^{-12}).

The column is chemically almost frozen. Between the deep boundary and the match, 17 decades of pressure, q_{H_2} moves from 0.19258 to 0.19264, q_{CH_4} from 1.072×10^{-4} to 1.025×10^{-4} and q_{NH_3} from 3.158×10^{-5} to 2.919×10^{-5} . What changes at all is water, which condenses, and a set of trace photochemical products at the very top. In an eddy-mixed column at $K_{zz} = 10^9$ that is what one expects, and it is the same statement as Sect. 11. makes from the other side — the eddy term homogenizes the column all the way to the match.

Hydrogen stays molecular to the top. q_{H} rises from 5.9×10^{-21} at the deep boundary to 6.5×10^{-6} at the match and 4.0×10^{-5} at the model top, four to five decades below q_{H_2} throughout: there is no level in this column at which atomic

hydrogen overtakes H_2 . The base EXHALE is handed is an H_2 base in that sense. It does not enter the wind that way in these runs — EXHALE's molecular chemistry is off, and q_{H_2} is read but inert (below) — which is worth stating rather than leaving implicit.

The composition is cold and reducing. At 520 K and 17 bar CO is already negligible, $q_{CO} = 8.1 \times 10^{-18}$, and CO_2 smaller still: carbon is CH_4 and nitrogen is NH_3 , the opposite of the CO/ N_2 -dominated deep chemistry of a hot Jupiter. Counting carriers at the match against the hydrogen-nucleus fraction of the gas, $2q_{H_2} + q_H = 0.38532$:

- carbon: CH_4 gives 2.661×10^{-4} per H, 99.7% of the carbon the profile's carrier columns hold, with C_2H_2 and HCN supplying the rest;
- nitrogen: NH_3 gives 7.576×10^{-5} per H, 96.3%, with N_2 at 3.4%;
- oxygen: the water that survived the cold trap gives 4.831×10^{-7} per H, 99.9%.

Those carrier sums come to 96.0, 96.0 and 97.6 per cent of the elemental columns $C/H = 2.778 \times 10^{-4}$, $N/H = 8.187 \times 10^{-5}$ and $O/H = 4.957 \times 10^{-7}$; the remainder sits in species the profile does not carry as columns, so the carrier accounting here is a decomposition of the elemental ratios and not an independent measurement of them.

This is the direct link to Sect. 11.5. The C and N that cool the wind and take its line down by $2.6\times$ are held in the column as CH_4 and NH_3 , both of which the solution keeps in the gas phase across the trap, so they cross the tropopause intact and arrive at the match at essentially their deep values. Oxygen, whose carrier is water, does not.

Photochemistry acts only in the top decades. From the tropopause to the match, HCN goes $1.3 \times 10^{-13} \rightarrow 6.5 \times 10^{-8}$, N_2 $9.2 \times 10^{-9} \rightarrow 5.2 \times 10^{-7}$, C_2H_2 $2.5 \times 10^{-13} \rightarrow 1.1 \times 10^{-7}$, CO $5.2 \times 10^{-16} \rightarrow 1.8 \times 10^{-10}$ and OH $4.5 \times 10^{-27} \rightarrow 7.1 \times 10^{-11}$ — several decades each, and all of them still trace. They do not move the elemental budget: the carrier accounting above puts every one of them below 4% of its element.

What EXHALE actually consumes from this file. Read in the code rather than inferred: the reader in `files_I0/lower_atmosphere_profile.f90` requires the columns `p`, `r`, `T`, `n_tot`, `rho`, `Kzz`, `q_H2`, `q_H` and `X_He` to be present, and `apply_lower_atmosphere_profile` (`input_read.f90`) then takes, *at the matching level only*, `T` into T_0 , `r` into R_0 , `q_H2` into q_{H2_base} , `X_He` into HeH , and `X_C`, `X_N`, `X_O` (and any other `X_{<El>}` the file carries) into the metal abundances, through the same door `metals.inp` uses. `p_match_bar` comes from the header. The `Kzz` column is the one thing read over the whole profile, interpolated onto the radial grid through the file's own `r` column (`eddy_diffusion_on_grid`); a scalar `He_Kzz`: in `input.inp` is then announced as inert. Everything else is *not* read: `n_tot` and `rho` are required by the schema but never consumed — the base density is n_0 of `input.inp` and the base particle count follows from T_0 , He/H and the elemental ratios — `q_H` is required and never consumed, the two flux columns `F_H` and `F_He` are not read by the Fortran at all (the closure driver reads them on the Python side), and none of the molecular columns beyond `q_H2` is read. And `q_H2` itself only acts when a molecular base is switched on: it reaches the base EOS through `h2_mixing_ratio_base`, which `comp_ntot_bc` calls only under `molecular_base`. In the runs of this memo that flag is off, and the resolved report confirms it — `mass_per_H = 9.37393` and `ntot_bc = 1.000116` are exactly $1 + 4He/H + \sum X_{El}A_{El}$ and $(1 + He/H + \sum X_{El})/(1 + He/H)$ with no H_2 term. So of the 23 columns, six numbers at one level plus one full column are what the wind is actually built from.

What the reservoir does to the column. The ladder arms of Sect. 11.6 each carry their own solved column, and raising the reservoir changes it in three measurable ways (Fig. 14d–f). The deep boundary cools, $520.3 \rightarrow 439.4 \rightarrow 387.9$ K at 16.73 bar for He/H = 2.09, 5.01 and 10.31, while the stratosphere stays at 185 K. H_2 is diluted by number, $q_{\text{H}_2} = 0.1926 \rightarrow 0.0907 \rightarrow 0.0462$ at the match, with C/H and N/H unchanged to four digits because they are stated per hydrogen nucleus. And the cold trap loosens: O/H at the match goes $4.96 \times 10^{-7} \rightarrow 6.43 \times 10^{-7} \rightarrow 9.09 \times 10^{-7}$, the column of Table 9. Oxygen is the only element the reservoir moves here, and it is the one whose carrier condenses.

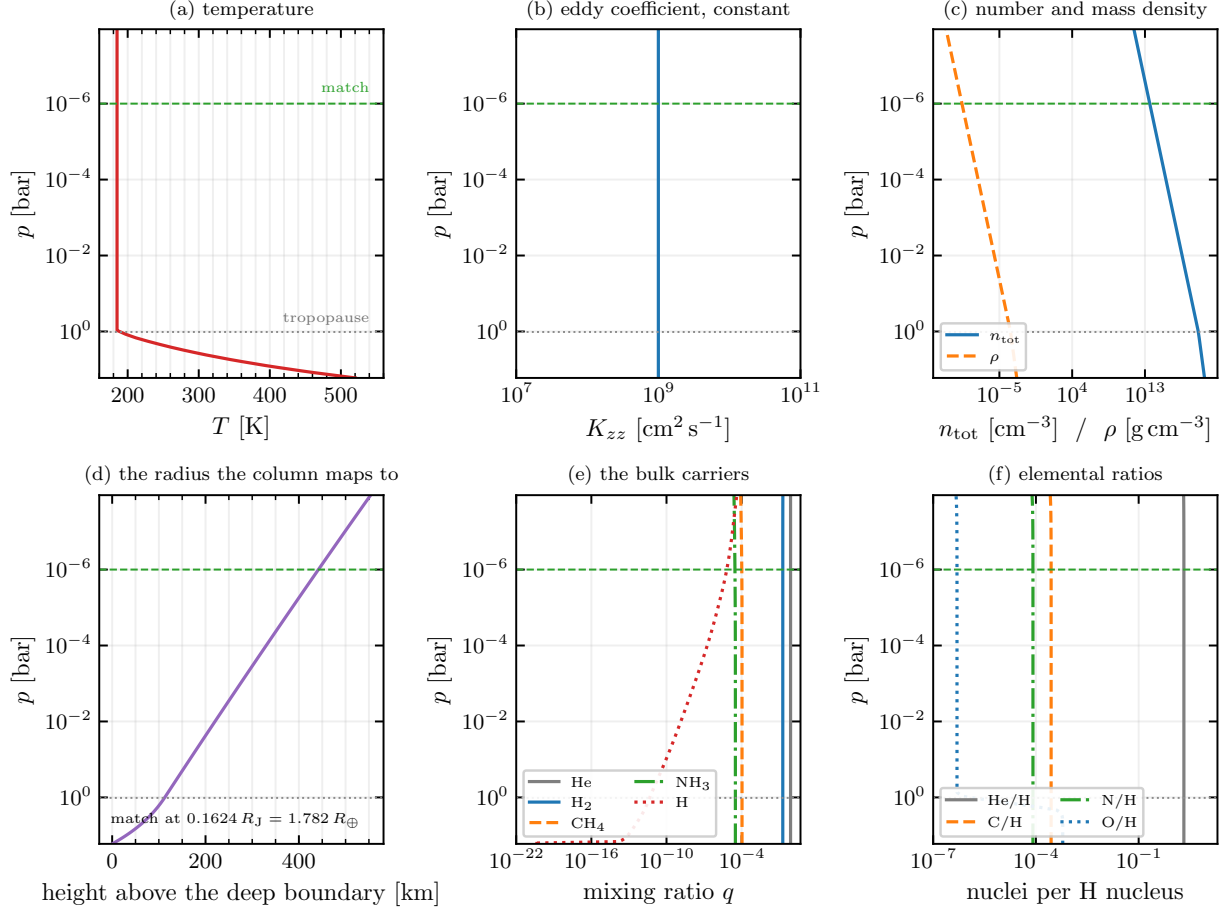


Figure 13: **The photochemical column, structure and bulk carriers.** (a) the solved temperature; (b) the eddy coefficient, constant at $10^9 \text{ cm}^2 \text{ s}^{-1}$ over all 102 levels; (c) number and mass density; (d) the radius the pressure grid maps to, which is what fixes the base radius of the wind; (e) the carriers that hold the bulk — helium reconstructed as $\text{He}/\text{H}(2q_{\text{H}_2} + q_{\text{H}})$, and atomic H four to five decades below H_2 everywhere; (f) the elemental ratios the profile hands over. The tropopause (1.031 bar, dotted) and the $1 \mu\text{bar}$ matching level (dashed) are marked on every panel. Fig. 18b,c shows the same climate and elemental ratios in the context of the closure; this figure is the file itself, column by column.

11.3 The cold trap the climate solve puts in

Solving the temperature rather than prescribing it puts a cold trap into the handoff, and the trap is where the climate solution says it is rather than where a closed form assumes it.

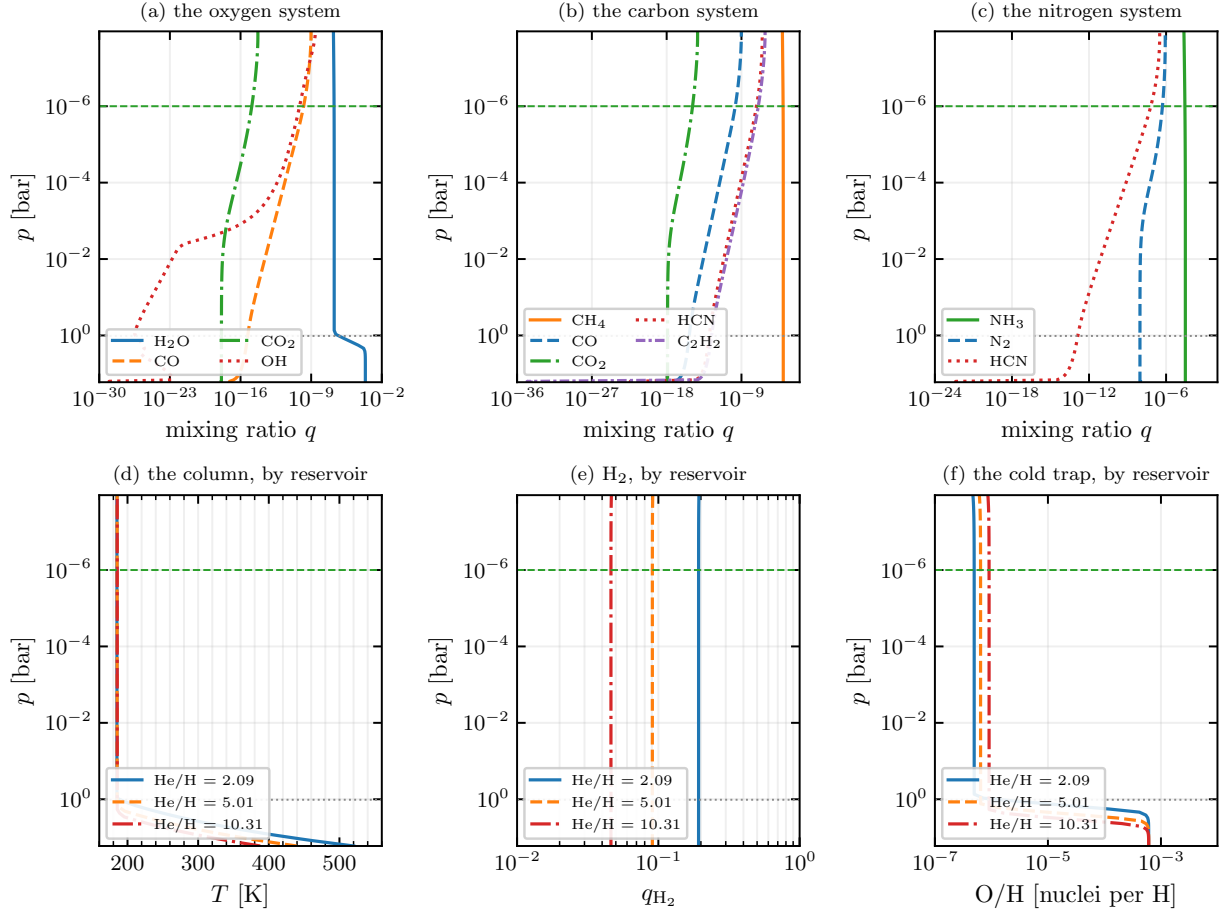


Figure 14: **The molecular systems, and what the reservoir does to them.** (a)–(c) the oxygen, carbon and nitrogen carriers of the same file: water condenses out at the tropopause while CH_4 and NH_3 cross it unchanged, and the photochemical products (HCN , C_2H_2 , N_2 , CO , OH) climb several decades in the top of the column while staying trace. (d)–(f) the same column solved at three reservoirs of the ladder of Table 9: the deep boundary cools as helium is added, H_2 is diluted by number, and the cold trap loosens, so oxygen is the one element the reservoir moves. Levels marked as in Fig. 13.

Table 6: The climate solve over its one free parameter, the deep boundary. $f_{\text{H}_2\text{O}}$ is the water mixing ratio above the tropopause, i.e. what crosses into the photochemical column. The stratospheric temperature is the robust part; the tropopause pressure is not, and the trapped water follows it, since at fixed T_{trop} the trapped ratio is $p_{\text{sat}}(T_{\text{trop}})/P_{\text{trop}}$.

deep boundary [bar]	T_{deep} [K]	P_{trop} [bar]	T_{trop} [K]	$f_{\text{H}_2\text{O}}$
1	242.4	0.488	186.0	3.33×10^{-7}
3	308.5	0.765	185.8	2.06×10^{-7}
10	439.9	0.974	185.4	1.49×10^{-7}
20 (adopted)	555.9	1.031	185.0	1.32×10^{-7}
30, 50	refused: the pseudoadiabat leaves the thermodynamic data			

T_{trop} moves by 1 K over a factor of 20 in the deep boundary, against the skin temperature $T_{\text{eq}}/2^{1/4} = 190$ K; P_{trop} moves by a factor of two, and the water moves with it. The sensitivity is measurable from the scan itself, since $f_{\text{H}_2\text{O}}P_{\text{trop}}$ is $p_{\text{sat}}(T_{\text{trop}})$: across the rows it gives 19% per K at 185 K, so the trapped water is conditional on the tropopause temperature at about a fifth per kelvin and on the tropopause pressure linearly.

Against the published estimate. Cherubim et al. (2026) quote $f_{\text{H}_2\text{O}} \sim 7$ ppm from a closed-form cold trap at a 0.1 bar tropopause with $T_{\text{skin}} = 194$ K. The climate solution gives 0.13–0.33 ppm at a tropopause of 0.5–1.0 bar: a factor 20–50 less water at a tropopause 5–10 $\times$ deeper. The saturation physics agrees — $p_{\text{sat}}(185 \text{ K})/1 \text{ bar}$ is 1.3×10^{-7} and $p_{\text{sat}}(194 \text{ K})/0.1 \text{ bar}$ about 5×10^{-6} , so each number follows from its own tropopause — and what differs is where the tropopause is put. Ours is a solved level in a helium-dominated column ($\mu = 3.6$ at $\text{He}/\text{H} = 2.09$), theirs an assumed one. This is reported as a result about the climate model and was not tuned toward the published number.

What the trap does to the handoff is a depletion with height, not a condensate sitting at the base: gas-phase oxygen falls from $\text{O}/\text{H} = 6.062 \times 10^{-4}$ at the deep boundary to 4.957×10^{-7} at the match, a factor 1223, while C/H (2.778×10^{-4}), N/H (8.187×10^{-5}) and He/H cross the tropopause unchanged — the carriers of carbon and nitrogen in this column are CH_4 and NH_3 (Sect. 11.2), and this solution keeps both of them in the gas phase across the trap (Figs. 18c and 14a–c).

11.4 The closure, and what it returns

Three arms were run, from the reference trial pair $(\Phi_{\text{H}}, \Phi_{\text{He}}) = (1.8, 1.3) \times 10^7 \text{ g s}^{-1}$ and from $0.3\times$ and $3\times$ it, all seeded from the same converged wind and all starting from the same $\text{He}/\text{H} = 2.09$ reservoir.

Table 7: The elemental-flux closure on LHS 1140 b. k is the iteration at which the 5% residual criterion was met. The equivalent width is re-measured here on this memo’s own red-pair window (10832.60–10834.20 Å vacuum), so it is directly comparable with the observed $1.108 \pm 0.030 \text{ \% Å}$; the same curves integrated over the whole synthesized band, which is how Update_EXHALE.md Sect. 79 records them, give 4.74–4.82 mÅ.

start	k	$F_{\text{H}} [\text{g s}^{-1}]$	$F_{\text{He}} [\text{g s}^{-1}]$	He/H at match	$\log_{10} \dot{M}$	red [%]	$EW [\text{\% Å}]$
1.0 $\times$	0	1.8143×10^7	1.3204×10^7	2.09235	7.500	1.699	0.421
0.3 $\times$	5	1.8203×10^7	1.3427×10^7	2.09235	7.500	1.721	0.426
3.0 $\times$	6	1.8212×10^7	1.3471×10^7	2.09236	7.500	1.725	0.427
observed	—	—	—	—	—	$1.254^{+0.22}_{-0.23}$	1.108 ± 0.030

The three arms agree on He/H to 5.5×10^{-6} , on $\log_{10} \dot{M}$ to better than the 0.005 the log prints, and on the fluxes to 0.4% and 2%, over a factor of ten in where they started: the closure is single-valued on this planet. The damping was never exercised — the residual fell on every iteration — but the $3\times$ arm used six of its eight allowed steps, so a start further than an order of magnitude out would need a larger iteration budget.

The composition is an output for the first time, and at this eddy coefficient it is the well-mixed value. He/H at the match comes back at 2.09235 against the 2.09 the reservoir was started from, a change of 1.1×10^{-3} , of which only 2×10^{-4} comes from the escape flux itself. The reason is measured: at $K_{\text{zz}} = 10^9 \text{ cm}^2 \text{ s}^{-1}$ eddy mixing homogenizes the column all the way to $1 \mu\text{bar}$, so a $1.8 \times 10^7 \text{ g s}^{-1}$ hydrogen escape imposed at the model top moves He/H by 1.8×10^{-2}

at that top and by 2×10^{-4} at the match. *It is the eddy coefficient, not the escape flux, that sets the composition at the matching level.* Closing the loop therefore does not retire the conditionality of Sect. 8.3; it relocates it, onto the same adopted K_{zz} .

Above the match the wind separates the two elements as strongly as before. The elemental fluxes carry $F_{\text{He}}/F_{\text{H}} = 0.73$ by mass against the base reservoir's 8.31, and He/H falls from 2.09 at the base to 0.183 at $30R_p$. The closure does not weaken that: it says the fractionation happens in the wind and not below the match. One caveat on the budget, stated rather than corrected: the diffusion operator's mass fraction is that of the H + He mixture and the trace metals ride inside ρ , so the measured F_{H} is hydrogen plus metals and overstates elemental hydrogen by 4.8×10^{-4} , two decades below the tolerance.

11.5 What the base metals do, and what the closed solution does not reproduce

The profile brings something the scalar base never had: the carbon, nitrogen and oxygen of the photochemical column. Isolating it takes a ladder of controls in which the scalar-base run is moved one key at a time onto the profile's own base state, every rung a converged solution (Update_EXHALE.md Sect. 78). Setting the base helium ratio to the profile's 2.0921 moves $\log_{10} \dot{M}$ by 0.01; setting the base temperature to the solved 185.05 K, from the 226 K the scalar runs used, moves it by less than 0.01 more — the base temperature is not the lever. Adding the base radius and molecular fraction moves it to 7.86. Adding the profile's C/H, N/H and O/H takes it to **7.46**, and the red-pair depth from 4.40% to **1.52%**: -0.40 dex in $\dot{M}$ and a $3.3\times$ shallower helium line, from metal-line cooling of a 185 K base at $\text{C}/\text{H} = 2.8 \times 10^{-4}$. The profile route reproduces that rung to every digit printed, which is the consistency check the two routes exist to give.

The consequence for the line is that the flux-closed solution does not reproduce the measured equivalent width.

At $K_{zz} = 10^9$ and a metal-free base, $\text{He}/\text{H} = 2.13$ gives $EW = 1.129\% \text{ \AA}$, which is why Table 2 puts the crossing at 2.09. The flux-closed solution sits at the same eddy coefficient and essentially the same reservoir, 2.09235, and gives $0.421\% \text{ \AA}$ — a factor 2.6 below the observed 1.108 ± 0.030 , and 23σ from it. The difference is entirely the base metals beneath it.

So closing the elemental fluxes and matching the equivalent width are two different constraints, and the solution presented at 2.09 satisfies the first only. Fixing the reservoir by the line, as Sects. 7. and 8.3 do, was done above a base carrying no metals; with the photochemical column beneath it the same reservoir absorbs $2.6\times$ less, so the composition that satisfies both conditions at once is larger than 2.09. Sect. 11.6 solves for it, at the adopted K_{zz} , and returns $\text{He}/\text{H} = 10.7$. So the lower boundary is an axis of its own, of the same standing as the assumed spectrum, the assumed mixing and composition itself: what is placed beneath the wind is not a free parameter once the lower atmosphere is solved, but is set by the elemental vector and the cold trap of the column below.

The mechanism: helium as a thermostat

Why the same reservoir absorbs differently over the two bases can be read off the solved structures directly. Table 8 measures the three solutions that bracket the question: the metal-free scalar base at $\text{He}/\text{H} = 2.13$, which reproduces the line (LHS1140b/exhale/heh2p13_diff_kzz1e9/); the flux-closed solution on the photochemical column at 2.09,

which does not (.../flux_closure/hi/k06/); and the closed solution at 10.31, the rung just below the crossing (.../flux_closure/heh10p3/k01/). All three run with H/He element diffusion active at $K_{zz} = 10^9 \text{ cm}^2 \text{ s}^{-1}$ — constant through the profile in the two profile runs, the scalar value in the third — and with EXHALE’s molecular chemistry off, so no molecular carrier appears in the wind. Mean mass per particle is $(n_{\text{H}} + 4n_{\text{He}})/(n_{\text{H}} + n_{\text{He}} + n_e)$; nuclei are counted as H I + H II and He I + He II + He III. The values are from the advection-corrected profiles; the reference arm of the closure (.../flux_closure/ref/k00/) reproduces the 2.09 column to under 1% in every entry.

Table 8: The two directions the thermostat runs, measured on the solved structures. Columns are the metal-free scalar base, the flux-closed solution on the photochemical column at the same reservoir, and the closed solution at 10.31. *EW* is on this memo’s red-pair window against the observed $1.108 \pm 0.030 \text{ \% } \text{\AA}$.

	scalar base	photochem base	photochem base
	no metals	C, N, O	C, N, O
reservoir He/H	2.13	2.09	10.31
T_{max} [K]	5330	3944	5869
$T(2R_p)$ [K]	4140	2979	5098
mean mass per particle at $2R_p$ [m_{H}]	2.00	1.51	2.53
$n(\text{H I})$ at $2R_p$ [cm^{-3}]	5.67×10^7	5.37×10^7	1.31×10^7
He/H by nuclei at $2R_p$	0.570	0.231	1.421
$\max n(2^3S)$ [cm^{-3}]	42.9	50.2	122.5
$\int n(2^3S) dr, 1-10R_p$ [$\text{cm}^{-3}R_p$]	65.4	40.5	96.9
of which $1-2R_p$	24.9	25.0	59.2
of which $2-10R_p$	40.5	15.5	37.7
<i>EW</i> [% $\text{\AA}$]	1.129	0.421	1.056

Adding the base metals cools the wind, and the line loses its outer column. With no metals, radiative cooling is 6–9% of the heating over $1.2-2R_p$ and the wind is limited by expansion rather than by radiation. With the column’s C, N and O beneath it, cooling reaches 71% of the heating at $1.2R_p$, of which 95% is metal-line cooling and the largest single contributor is neutral C I; T_{max} falls from 5330 to 3944 K. The metastable peak does not fall with it — it rises, $42.9 \rightarrow 50.2 \text{ cm}^{-3}$ — and neither does the column inside $2R_p$, which is the same to 0.4% (24.9 against 25.0). What is lost is entirely outside $2R_p$: $40.5 \rightarrow 15.5 \text{ cm}^{-3}R_p$, a factor 2.6, matching the factor the equivalent width falls by ($1.129 \rightarrow 0.421 \text{ \% } \text{\AA}$). The cooler wind has the smaller scale height, so it does not carry the metastable population out to where it adds absorbing area.

Raising the reservoir reheats it, by two terms that are both measured. From 2.09 to 10.31 on the same base, T_{max} goes $3944 \rightarrow 5869 \text{ K}$. Two things change together. First, there are fewer particles to share the deposited energy: the mean mass per particle at $2R_p$ rises from 1.51 to $2.53 m_{\text{H}}$, a factor 1.68, and the heating rate per particle at $2R_p$ rises by 1.88, while $T(2R_p)$ rises by 1.71. The three factors agreeing to within 10% is a consistency check on that reading, not a proof of it — the cooling term is not held fixed between the two runs. Second, the coolant is diluted along with the hydrogen it is referred to: the profile states C, N and O per hydrogen nucleus (schema column $X_*[\text{El/H nuclei}]$, and in the code $n_X = \text{melem_ab}(e)n_{\text{H}}$), and the two profiles carry the same $\text{C/H} = 2.778 \times 10^{-4}$ to four digits, so when

$n(\text{H I})$ at $2R_p$ falls by 4.1 the carbon density falls with it by the same factor. The volumetric metal cooling nonetheless rises, $3.79 \rightarrow 6.05 \times 10^{-10} \text{ erg cm}^{-3} \text{ s}^{-1}$ at $1.5R_p$, because the cooling function is steeply temperature dependent; what the dilution does is let the gas settle at the higher temperature at which that happens.

What the metastable balance does with that. The 2^3S population in these solutions is fed almost entirely by recombination and drained by Penning ionization on neutral hydrogen and by electron collisional de-excitation. Reconstructing the residual EXHALE solves — source $n_e n_{\text{He II}} \alpha_{2^3S}$ plus collisional excitation from the singlet ground state, against radiative decay A_{31} , photoionization, Penning on H I at the Taylor et al. (2025) rate, electron collisional de-excitation to 2^1S and 2^1P , and electron collisional ionization (`ion_residual_core.f90`, `tr_triplet_row`) reproduces the tabulated $n(2^3S)$ to five digits, so the budget can be apportioned. At the metastable peak, collisional excitation supplies under 0.05% of the source, and radiative decay, photoionization and collisional ionization together take under 0.1% of the sink. What is left is: at $\text{He}/\text{H} = 2.09$, Penning 89% and electron de-excitation 11% of the loss; at 10.31, Penning 60% and electron de-excitation 40%. Between the two, the source rate rises by 1.55 (more He II and more electrons in the hotter gas, against a recombination coefficient that falls with temperature) and the metastable lifetime lengthens by 1.57, from 0.47 to 0.74 s: hydrogen has been thinned by 3.2 at the peak, so Penning removes the level more slowly, and the faster electron de-excitation of the hotter gas takes back only part of it. The product, 2.44, is the peak ratio $50.2 \rightarrow 122.5$. *There is no charge-exchange term on the metastable in this code* — the He/H charge exchange acts on ground-state He and He^+ — so it is not part of this balance.

Read as an interpretation, the thermal structure of the 10.31 solution ($T_{\text{max}} = 5869 \text{ K}$, column 96.9) has recovered and slightly passed that of the metal-free 2.09 solution (5330 K, 65.4), which is why the equivalent width comes back to the measured value there. The ladder therefore looks less like a determination of composition than like a thermal trade between the metal content of the base and the helium content above it, and one line cannot separate the two. Helium in these solutions is not only the absorber; it is also the thermostat that sets how far the absorber is carried.

What the wind is made of, and how ionized it is. Fig. 15 carries the elemental fractions and the ionization state of the same three solutions, on the same advection-corrected profiles as Table 8. Two things follow from it that the thermostat argument rests on.

Diffusion keeps the wind far less helium-rich than the reservoir. On both branches the helium fraction falls steeply through the line-forming region and then flattens: $\text{He}/\text{H} = 2.09$ at the base gives 0.231 by nuclei at $2R_p$ and 0.186 beyond $5R_p$ on this arm (0.183 on the reference arm quoted in Sect. 11.); 10.31 gives 1.421 and 1.156. That is a factor 9.1 and 8.9 between the reservoir and the wind, in a solution where the reservoir is an output of the closure rather than a choice. The consequence is worth stating in the composition's own terms: the flux-closed solution at 2.09 — a reservoir that is 68 per cent helium by number — has a *hydrogen-dominated* line-forming region, $n_{\text{H}}/n_{\text{tot}} = 0.81$ against $n_{\text{He}}/n_{\text{tot}} = 0.19$ at $2R_p$. At 10.31 helium still leads there, 0.59 against 0.41, but by a factor 7 less than the reservoir states. A composition quoted for this planet therefore has to say at which level it is quoted; the reservoir and the absorbing layer are not the same number, and the memo's crossings are reservoir values.

The hotter solution is also the more ionized one, and it is not shielding. At $2R_p$, $x(\text{H II}) = 0.031$ on the photochemical base at 2.09 and 0.106 at 10.31 — a factor 3.4, in the run whose neutral hydrogen density is 4.1 times lower. The obvious reading, that thinner hydrogen shields itself less, is contradicted by the outputs: the H I column above $2R_p$

gives $\tau(13.6\text{eV}) = 0.23$ and 0.055 , so both are optically thin there, and the H I photoelectric heating per neutral hydrogen atom is the same in the two, 1.23 against $1.26 \times 10^{-18} \text{ erg s}^{-1}$ per atom. The recombination sink barely moves either: $\alpha_B(T)n_e$ is 2.35 against $2.13 \times 10^{-6} \text{ s}^{-1}$. What grows is the ionization of hydrogen by *helium*. Measured on the equilibrium state at $2R_p$, the two channels that can be computed in closed form from the outputs are Penning ionization $\text{He}(2^3S)+\text{H}$, which supplies 31% of what case-B recombination removes at 2.09 and 24% at 10.31 — a falling share but a rate per neutral hydrogen atom 2.8 times higher — and the charge exchange He^++H , at 2.3% and 1.5%. The rest, 67% and 75%, is photoionization, and inside it the helium-fed parts are what scale: the He-recombination-radiation channel (He II recombining to ground, whose $\geq 24.6 \text{ eV}$ continuum ionizes H) carries 6.4 times more heating per neutral hydrogen atom in the 10.31 solution, and the energy deposited by He I photoionization per neutral hydrogen atom, part of which secondary ionization turns into hydrogen ionizations, is 7.0 times higher. *What could not be settled from the outputs* is the split inside that 67–75%: the individual rates — direct Γ_{HI} , the He-recombination-driven addition to it, and the secondary-ionization yield — are not written to any output file, and the factors quoted here are read off the heating columns on the assumption that the mean photoelectron energy is similar between the two runs. The one channel that can be excluded outright is $n = 2$ photoionization: `heat_Hpe[excitedH]` is exactly zero at every cell of all three runs. Helium ionization itself tracks the temperature more simply — $x(\text{He II})$ is 0.053, 0.065 and 0.079 at $2R_p$ across the three, and $x(\text{He III})$ stays below 2×10^{-4} everywhere inside $3R_p$.

Fig. 11 shows the same kind of structure for the single well-mixed solution that reproduces the line with no diffusion and no base metals; Fig. 15 is the set the lower-boundary axis is built on, so the two are complementary rather than alternative views.

11.6 The composition the line needs above a photochemical base

Sect. 11.5 left the composition that closes the fluxes *and* reproduces the line unsolved. This section solves for it on the composition axis, at the adopted eddy coefficient. What comes out is the answer on the solved-column branch alone: it is the counterpart of the 2.09 that the same line returns above a prescribed metal-free base (Sect. 8.3), not a replacement for it, and the two differ by the base and not by the mixing, which is $K_{zz} = 10^9$ with diffusion active in both. The setup: the closure driver is run at six further reservoirs, each arm started from the converged flux and the converged wind of the rung below it, so every arm begins near its own fixed point. Everything else is the configuration of Sect. 11. — Photochem with the `clima` step solved, $K_{zz} = 10^9$, a $1 \mu\text{bar}$ match, the GJ 1132 proxy spectrum, a 5% residual tolerance and $k_{\text{max}} = 8$. The runs are `LHS1140b/exhale/flux_closure/heh{3,5,8,9p7,10p3,12}/`.

Every arm returns a matching-level He/H equal to its reservoir to four digits, for the reason Sect. 11. measured at 2.09: at $K_{zz} = 10^9$ the eddy-mixed column reaches the microbar match, so the escape flux does not set the composition there. What the ladder adds is the line.

The crossing. The arms at 10.31 and 12.01 straddle the measured equivalent width, so the answer is interpolated inside a bracket rather than extrapolated off the end of a scan:

$$\text{He/H} = 10.7 \quad (10.5\text{--}10.9 \text{ on the } EW \text{ } 1\sigma \text{ interval}), \quad (7)$$

by log–log interpolation of the seven scanned points solved with Brent’s method; the range carries the measurement error alone and none of the model uncertainty. *No run was made at 10.7 itself* — the value is read off the bracket,

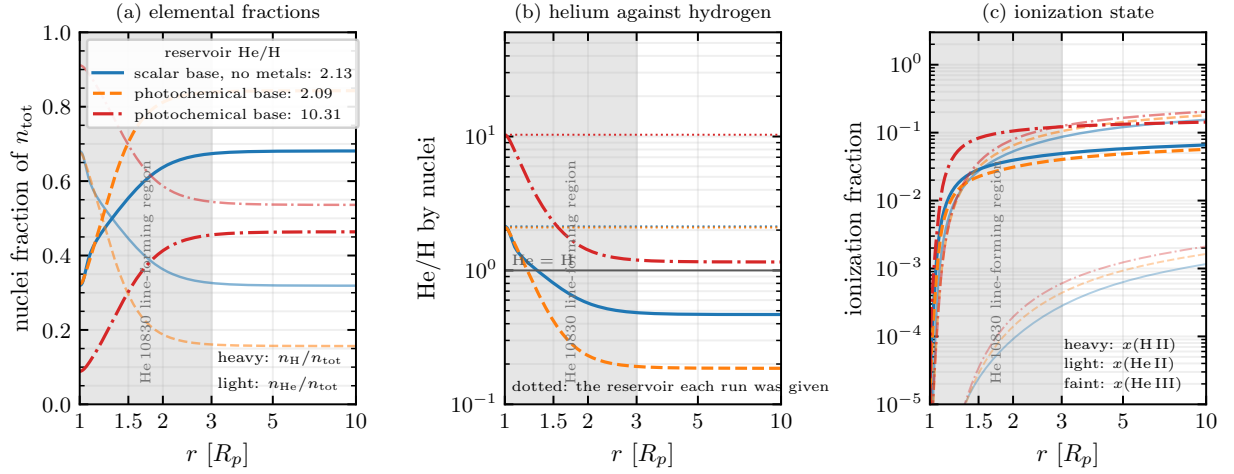


Figure 15: **Composition and ionization against radius**, for the three solutions of Table 8, from the same advection-corrected `Ion_species_adv.txt` profiles. **(a)** the hydrogen and helium nuclei fractions of n_{tot} ; **(b)** He/H by nuclei, with the reservoir each run was given as a dotted line, so the vertical drop from dotted line to curve is what element diffusion does; **(c)** the ionization fractions. n_{tot} follows the code’s `calc_ntot` convention — one particle per species, H I, H II, He I, He II, He III and every metal ion stage, electrons excluded, and He 2^3S not added because it is an excited level inside He I — so (a) does not sum to one on the photochemical base. The metals it misses are negligible: their largest share of n_{tot} anywhere in $1\text{--}30R_p$ is 3.0×10^{-4} at He/H = 2.09 and 1.7×10^{-4} at 10.31, and zero on the metal-free base. Nuclei are counted as H I + H II and He I + He II + He III. The 10.31 run is the rung nearest the crossing at He/H = 10.7; no run was made at 10.7 itself (Sect. 11.6). The He 10830 line-forming region $1\text{--}3R_p$ is shaded.

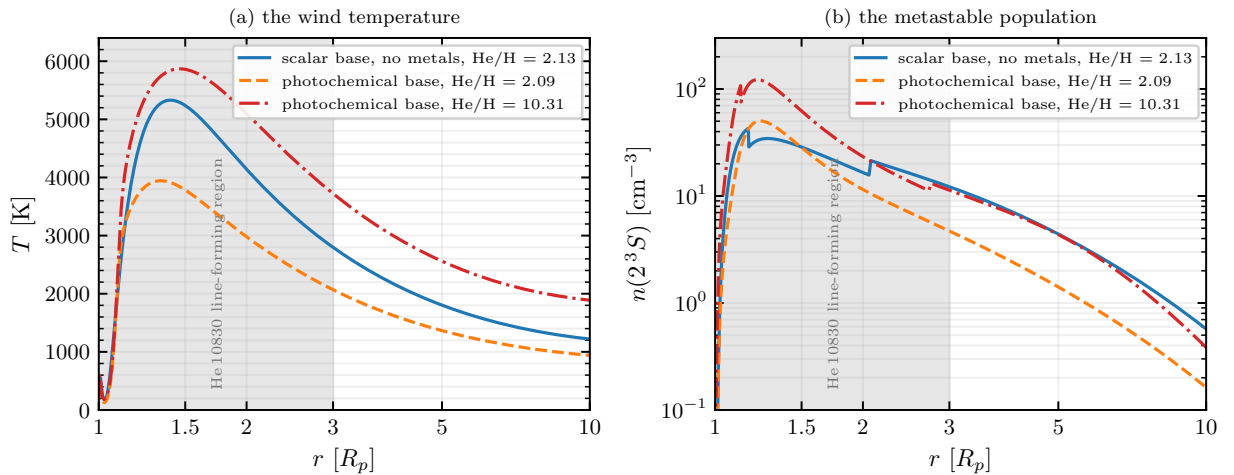


Figure 16: The thermostat, on the three solutions of Table 8. **(a)** temperature and **(b)** the metastable density, against radius, with the He 10830 line-forming region $1\text{--}3R_p$ shaded. Adding the photochemical column’s C, N and O at fixed reservoir (solid to dashed) cools the wind and strips the metastable population outside $2R_p$ while leaving the peak alone; raising the reservoir on that same base (dashed to dot-dashed) reheats it past where it started and carries the population back out.

Table 9: The reservoir ladder under flux closure. One row per arm, at the iteration k where the 5% residual criterion was met; fluxes in 10^7 g s^{-1} , the cold-trap oxygen in 10^{-7} nuclei per H at the matching level. The equivalent width is on this memo's red-pair window, directly comparable with the observed $1.108 \pm 0.030 \text{ \% } \text{\AA}$.

reservoir	k	F_{H}	F_{He}	He/H at match	$\log_{10} \dot{M}$	red [%]	FWHM [$\text{\AA}$]	$EW \text{ [% } \text{\AA}]$	EW/obs	O/H trap
2.09 (Sect. 11.)	0	1.814	1.320	2.0924	7.500	1.699	0.2326	0.4206	0.380	4.95
3.0	3	1.369	1.491	3.0034	7.460	2.122	0.2386	0.5388	0.486	5.54
5.0	4	0.880	1.692	5.0057	7.410	2.795	0.2497	0.7433	0.671	6.43
8.0	4	0.572	1.894	8.0090	7.390	3.388	0.2608	0.9437	0.852	7.94
9.7	3	0.477	2.043	9.7109	7.400	3.660	0.2664	1.0403	0.939	8.78
10.3	1	0.450	2.069	10.3116	7.400	3.696	0.2674	1.0563	0.953	9.09
12.0	4	0.391	2.238	12.0135	7.420	4.630	0.2654	1.3126	1.185	10.66
observed	—	—	—	—	—	$1.254^{+0.22}_{-0.23}$	0.841	1.108 ± 0.030	1	—

and the bracket has a real width: the equivalent width grows sublinearly with the reservoir, at a local power-law slope of 0.69 at the bottom of the ladder and 0.51 between 5 and 10, but the two rungs that straddle the measurement do not follow that trend, giving 0.25 between 9.71 and 10.31 and 1.42 between 10.31 and 12.01. That mismatch of local slopes, not the interpolation error, is what the quoted width stands on, and it is not resolved here. What the sublinearity does settle is that the equivalent width does *not* saturate before the measurement is reached, which the helium-rich end of the metal-free ladder of Sect. 8.3 had made a live possibility.

At the crossing the escape rate is $\log_{10} \dot{M} = 7.40\text{--}7.42$, i.e. $2.5\text{--}2.6 \times 10^7 \text{ g s}^{-1}$, a factor 8 below the $2.03^{+0.58}_{-0.67} \times 10^8 \text{ g s}^{-1}$ the p-winds retrieval imposes, and the elemental fluxes are $F_{\text{H}} \simeq 4.3 \times 10^6$ and $F_{\text{He}} \simeq 2.1 \times 10^7 \text{ g s}^{-1}$.

Two statements follow, and they are worth separating.

- **The closure by itself does not select a reservoir on this planet.** Every rung of Table 9 satisfies it, at residual flux spreads of 0.4–2.6% against the 5% tolerance. What selects 10.7 is the line, exactly as in Sects. 7. and 8.3. What the closure contributes is that the value is now consistent with a photochemical lower atmosphere carrying its own C, N and O, rather than resting on a bare H/He base — and the price of that consistency is a helium fraction of 91 per cent by number. Read against the 2.09 the same line returns above a metal-free base, the pair measures the thermal trade of Sect. 11.5 rather than two attempts at one number.
- **The escape rate is flat along the whole ladder**, 7.39 to 7.50 in $\log_{10} \dot{M}$ over a factor 5.7 in reservoir. So here too the composition is fixed by the line and not by the wind, and the $\dot{M}$ this memo quotes does not depend on where in the ladder the crossing lands.

The wind residual the JFNK finish reaches degrades as the reservoir grows — the line search stalls on the base contact mode and returns its best iterate — so the target was raised arm by arm and the achieved values recorded rather than smoothed over: $\|R\| = 9.8 \times 10^{-5}$ at 2.09 rising to $2.4\text{--}3.4 \times 10^{-4}$ at 8 and above, all at $\text{info} = 0$. The test the closure actually applies is not that residual but the radial spread of the elemental flux over the steady window, which is what the 0.4–2.6% above are.

What the crossing does not fix, and what it inherits. The line is still too narrow: FWHM $0.267 \text{ \AA}$ at the crossing against the measured 0.841 , unchanged from Sect. 8.3 and from the diffusion-off solutions, so the modelled red depth of 3.7 per cent overshoots the measured 1.254 per cent by the same factor the width falls short — which is why the composition here is inferred from the equivalent width and not from the depth. And the winds these numbers come from carry the caveat of Sect. 12. in full: they are subsonic to the $30R_p$ domain edge, their mass flux is set at the outer boundary rather than at a critical point, and the critical region is transitional, so the $\log_{10} \dot{M} = 7.40\text{--}7.42$ quoted above is *unvalidated as a continuum solution*. The He 10830 layer that fixes 10.7 is not: it sits at $\text{Kn} \leq 0.021$.

11.7 The 2025 non-detection

Cherubim et al. detect the line in 2024 and not in 2025, at a limit of 0.6 per cent in depth. The grid below asks what fraction of the fiducial XUV reproduces that. The spectrum is the fiducial GJ 1132 proxy with its flux column multiplied by the stated factor (sed/lhs1140_sed_gj1132_at_b_xuv*.txt; read_sed recomputes L_X and L_{EUV} from the file, so the luminosity lines of input.inp are provenance only). The lower atmosphere is held fixed in every case — the closure loop is not re-run, and neither is the chemistry that made the profile — so the grid isolates the wind’s response to the irradiation and nothing else.

It is run from both models that reproduce the 2024 equivalent width, because neither is the other: the scalar-base solution of Sect. 8.3 at $\text{He}/\text{H} = 2.13$ ($EW = 1.129$), and the flux-closed solution of Table 9 at $\text{He}/\text{H} = 9.71$ ($EW = 1.040$). Runs LHS1140b/exhale/xuv*_heh2p13/ and xuv*_closure9p7/.

Table 10: The He 10830 line against the XUV scaling, from the two models that reproduce the 2024 measurement. Bold marks the row at which each model reaches the 0.6 per cent depth limit of the 2025 epoch.

$F_{\text{XUV}} / \text{fiducial}$	scalar base, $\text{He}/\text{H} = 2.13$			flux-closed, $\text{He}/\text{H} = 9.71$		
	red [%]	$EW [\% \text{ \AA}]$	$\log_{10} \dot{M}$	red [%]	$EW [\% \text{ \AA}]$	$\log_{10} \dot{M}$
1.00	4.396	1.129	7.80	3.660	1.040	7.40
0.33	0.778	0.180	7.38	0.602	0.153	6.74
0.30	0.591	0.136	7.35	—	—	—
0.25	0.334	0.077	7.32	0.511	0.129	6.55
0.20	0.034	0.009	7.23	0.361	0.090	6.40
0.15	0.007	0.002	7.14	0.256	0.063	6.22
0.10	0.003	0.001	6.93	0.165	0.040	5.97
0.01	0.0003	0.000	6.10	0.190	0.055	5.51

The 0.6 per cent limit is met at about 0.3 of the fiducial XUV in both models, and that agreement is the useful part of the result: the two disagree by a factor 4.6 in reservoir and a factor 2.5 in escape rate, and still put the non-detection at the same place. In the scalar model the depth crosses 0.6 between 0.30 (0.591 per cent) and 0.33 (0.778); in the flux-closed model it is 0.602 at 0.33 and 0.511 at 0.25, so the crossing sits just under 0.33. On the paper’s fiducial $F_{\text{XUV}} = 33 \text{ erg cm}^{-2} \text{ s}^{-1}$, the 2025 epoch would need about $10 \text{ erg cm}^{-2} \text{ s}^{-1}$ or less.

Two caveats belong with that number. The decline is far steeper in the scalar model than in the closed one — a factor 200 in depth between 0.30 and 0.20 against a factor 1.6 — so the *shape* of the XUV dependence is a property of the model rather than a result; only the crossing agrees. And the lowest closed-model point, at 0.01, is not monotonic with the 0.10 one (0.190 against 0.165 per cent) on a wind that has fallen to $\log_{10} \dot{M} = 5.5$; that solution is not to be read as a limit on how faint the line can get.

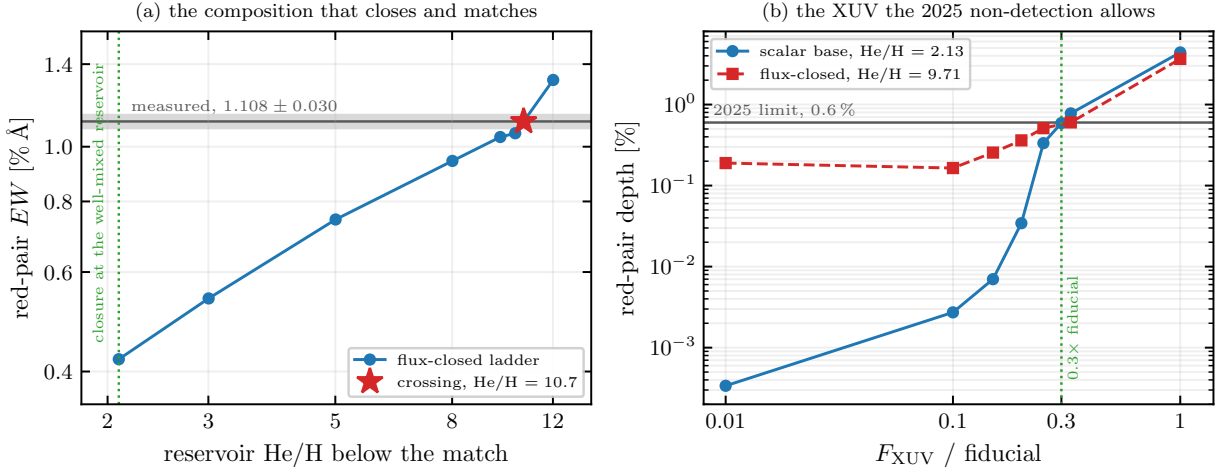


Figure 17: **The two constraints, and the epoch.** (a) The red-pair equivalent width of the flux-closed solution against the reservoir He/H it was closed at (Table 9), with the measured 1.108 ± 0.030 % Å as the band and the interpolated crossing at He/H = 10.7 as the star; the dotted line is the reservoir at which Sect. 11. closed the loop, five times lower. Every point on the curve satisfies the flux-closure condition, so it is the band and not the closure that picks a reservoir. (b) The red-pair depth against the XUV scaling (Table 10), from the scalar-base and flux-closed models that both reproduce the 2024 equivalent width. Both fall below the 0.6 per cent limit of the 2025 non-detection at about 0.3 of the fiducial irradiation, while the slope with which they get there differs by two orders of magnitude.

11.8 The width, unmoved again

Measured the way Sect. 8.2 measures it — convolve the solved profile with a Gaussian and ask which one reproduces the observed width — the flux-closed solution asks for 22.39 km s^{-1} FWHM ($\sigma = 9.51 \text{ km s}^{-1}$), against 22.30 in the well-mixed limit, 22.31 with diffusion at the adopted K_{zz} , and 22.08 on the $5.1 \times$ brighter proxy. The three arms agree to 0.01 km s^{-1} among themselves. As solved, the red-pair FWHM is 0.233 Å against the measured 0.841 .

The lower-boundary axis therefore behaves like the other three: it moves the composition inference and the escape rate, and it does not move the width. It differs from them in one respect worth noting. Supplied as one Gaussian of 22.4 km s^{-1} , the closed solution reaches the observed width at a red depth of 0.469% against the observed $1.254^{+0.22}_{-0.23}$ — too *shallow*, where every earlier solution in this memo was matched in equivalent width first and therefore landed inside the observed depth interval. That is the same statement as Sect. 8.4 makes about the paper’s own p-winds fit: a solution short of absorbing power cannot be convolved into agreement, because broadening conserves equivalent width.

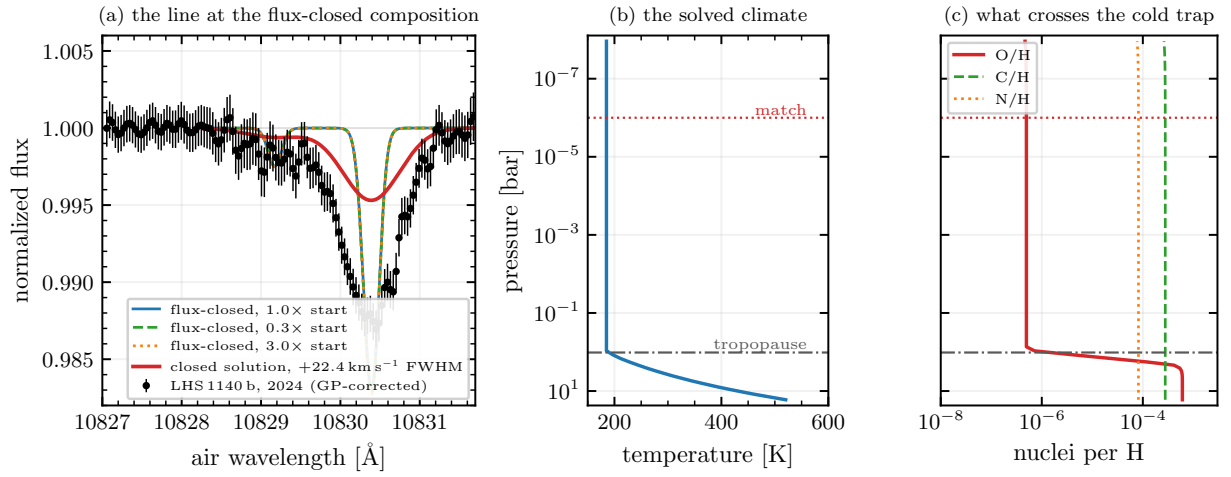


Figure 18: **The flux-closed solution and the column it stands on.** (a) The three closure arms — started a factor of ten apart in the trial escape flux — are indistinguishable, and all three fall well short of the measured line in absorbing power; the thick curve is the reference arm convolved with the 22.4 km s^{-1} FWHM Gaussian that reproduces the observed width, which then lands below the observed depth because the equivalent width was already short. Model curves are shifted by the fitted $v_{\text{wind}} = +2.26 \text{ km s}^{-1}$, as elsewhere in this memo. (b) The solved climate of the column handed over, with the tropopause at 1.031 bar and the $1 \mu\text{bar}$ matching level marked; the stratosphere sits at the skin temperature, 185 K. (c) The elemental abundances the profile carries. Oxygen is cut by a factor 1223 at the tropopause by the water cold trap, while carbon and nitrogen cross it unchanged — and it is those two, not the helium, that make this solution’s line $2.6\times$ weaker than the metal-free base at the same reservoir (Sect. 11.5). Panels (b) and (c) place the column in the closure; the file itself, every column of it, is Figs. 13 and 14.

12. Is the wind collisional where it is set?

Everything above reads the solutions as continuum solutions. A hydrodynamic wind is one only where the gas is collisional on the scale over which the flow varies, and above all through its critical point, which is where the topology — and with it the mass flux — is fixed. This section measures that. The tool is `src/utls/collisional_validity.py`; it reads an existing run directory and changes nothing, and the definitions and full tables are in `docs/collisional_validity.md` (record: `docs/Update_EXHALE.md` Sect. 81). The numbers below were measured on 2026-08-27 on the outputs the rest of this memo uses; nothing was re-run.

12.1 What is measured

The collisions. The four momentum-transfer limits are the ones the binary element-diffusion operator of Sect. 8.3 already uses, in the same Chapman–Enskog first approximation, so a mean free path quoted here and a diffusion coefficient used by the wind cannot drift apart: rigid-sphere neutral–neutral, non-resonant ion–neutral with the induced-dipole and rigid-core frictions added, and screened Coulomb with the Spitzer $\ln \Lambda$ for ion–ion and ion–electron. The step from a binary diffusion coefficient to a collision frequency is the definition of that coefficient itself, $\nu_{st} = n_t kT / (\mu_{st} \hat{D}_{st})$ with $\hat{D} = nD$, in which the total carrier density cancels. Species are H I, H II, He I, He II, He III and e , plus the molecular carriers where a run tracks them; He I 2^3S is a level inside the He I column and is not counted twice, and metals are dropped as trace ($< 10^{-3}$ of the particles) except in the electron budget. Two channels are deliberately absent — resonant charge exchange and electron–neutral momentum transfer — and both lengthen the mean free path, so the verdict they produce is the conservative one.

The numbers. $\lambda_s = \bar{v}_s / \nu_s$ with the harmonic sum over partners; the local structure scale is $L = \min(H_p, L_v, r)$ with $H_p = |d \ln p / dr|^{-1}$ measured from the solution and $L_v = |v / (dv/dr)|$ admitted only above Mach 0.01, below which it is the distance to a stagnation point rather than a structure scale; and $\text{Kn}_s = \lambda_s / L$. The *exobase* is the first crossing of $\text{Kn}_{\text{bulk}} = 1$ from below, never extrapolated beyond the domain. The *critical point* is the sonic point taken with the code’s own sound speed, $c_s = \sqrt{\gamma p / \rho}$ at $\gamma = 5/3$, and not a re-definition. The *critical region* runs from the peak of volumetric heating to the sonic point. “Collisional” means $\max \text{Kn} < 0.1$ across that region, over the bulk and over every species carrying at least 10^{-3} of the mass; where it fails the solution is reported **unvalidated**, not overestimated, because which way a collisionless critical region moves the answer is not something the continuum solution that assumed it away can decide.

12.2 The result

LHS 1140 b’s wind never reaches its critical point inside the computational domain. All three solutions stay subsonic out to $29 R_p$, at $\mathcal{M}_{\text{max}} = 0.54\text{--}0.57$. A subsonic solution is not fixed by a critical point: its mass flux is set by the outer boundary condition — and at that boundary Kn_{bulk} is already 1.3–2.3, with the exobase at $19.8\text{--}26.7 R_p$, inside the domain. The continuum equations were therefore integrated through, and closed at, a region in which they do not hold. The escape rates quoted throughout this memo, $\log_{10} \dot{M} \simeq 7.5\text{--}7.8$, are **unvalidated in the collisional sense**.

Table 11: Collisional validity of three representative LHS 1140 b solutions and of an HD 209458 b control, read from the advection-corrected profiles. Radii in R_p ; the LHS 1140 b domain ends at $29.05 R_p$, the HD 209458 b one at $4.15 R_p$. “Max Kn” is over the critical region, i.e. from peak heating to the sonic point, or to the outer boundary where there is no sonic point.

case	critical point	exobase	Kn = 0.1 at	max Kn	verdict
heh0p55 (no diffusion)	none, $\mathcal{M}_{\max} = 0.540$	25.22	6.42	2.2	unvalidated
heh2p13_diff_kzz1e9	none, $\mathcal{M}_{\max} = 0.565$	26.66	6.79	1.9	unvalidated
flux_closure/hi/k06	none, $\mathcal{M}_{\max} = 0.550$	19.76	5.53	3.0	unvalidated
HD 209458 b control	4.085	above 4.15	—	0.021	validated

Against the p-winds retrieval, whose isothermal Parker solution places its sonic point at $8\text{--}9.5 R_p$; that radius lies *inside* the exobase measured here, but Kn_{bulk} there is $0.14\text{--}0.30$, two to three times the threshold. So the finding is neither “the sonic point is above the exobase” nor “the critical region is collisional”: the critical region of LHS 1140 b is *transitional*. The scale separation the continuum equations need has been lost by the radius at which either code sets the mass flux, while the gas is not yet free molecular.

It is the neutral hydrogen that decides it, and the line does not care. In the critical region of the well-mixed solution the species split is H I 2.2, He I 1.6, H II 3.6×10^{-5} , He II 1.8×10^{-5} , He III 4.7×10^{-6} , e 6.5×10^{-4} (Fig. 19b). The neutrals carry the mass and the long mean free path; the charged species are Coulomb-held. That matters for reading the rest of this memo, because the He 10830 metastable population forms at $1.1\text{--}3 R_p$, where Kn_{bulk} is 1.2×10^{-4} at $1.1 R_p$, $0.004\text{--}0.006$ at $2 R_p$ and $0.013\text{--}0.021$ at $3 R_p$ in every case here. **The line-forming layer is collisional by two orders of magnitude.** What is unvalidated is the wind solution’s mass flux, not the absorption Sects. 7.–8. measure, and not the 22 km s^{-1} of missing width, which is asked for at radii where the continuum description is secure.

The single-temperature energy equation is not what fails. Electron–ion energy coupling is fast wherever it matters: the largest $\tau_{ei}^E/\tau_{\text{heat}}$ over the critical region is 0.030, 0.023 and 0.018 in the three LHS 1140 b solutions and 8.3×10^{-5} in the control. Momentum coupling times are likewise short compared with the flow time — $\tau_{\text{HI}} = 924 \text{ s}$ at $8 R_p$ against a flow time of $1.9 \times 10^5 \text{ s}$ — which is not a contradiction but a different test: a strongly subsonic flow gives hundreds of collisions per flow time even at $\text{Kn} \sim 1$, and it is the Knudsen number that decides continuum validity, because the terms the closure drops (viscous stress, heat flux, the non-Maxwellian tail) scale with the gradient over a mean free path rather than with the collision count.

The kinetic scale, and why it is not a bound. Jeans escape through the measured exobase (Chamberlain & Hunten 1987, eq. 7.2.5) gives $1.0\text{--}2.0 \times 10^7 \text{ g s}^{-1}$, a factor about three below the continuum $\dot{M}$. That is a *scale*, not a bound. Jeans escape bounds a continuum rate from below only when the exobase is well bound, and here $\lambda_J(\text{H}) = 0.78\text{--}1.45$: the exobase is barely gravitationally bound, the Jeans integral is most of the Maxwellian rather than its escaping tail, and the atmosphere is in blow-off. What the number does say is that the continuum result sits within a factor of a few of

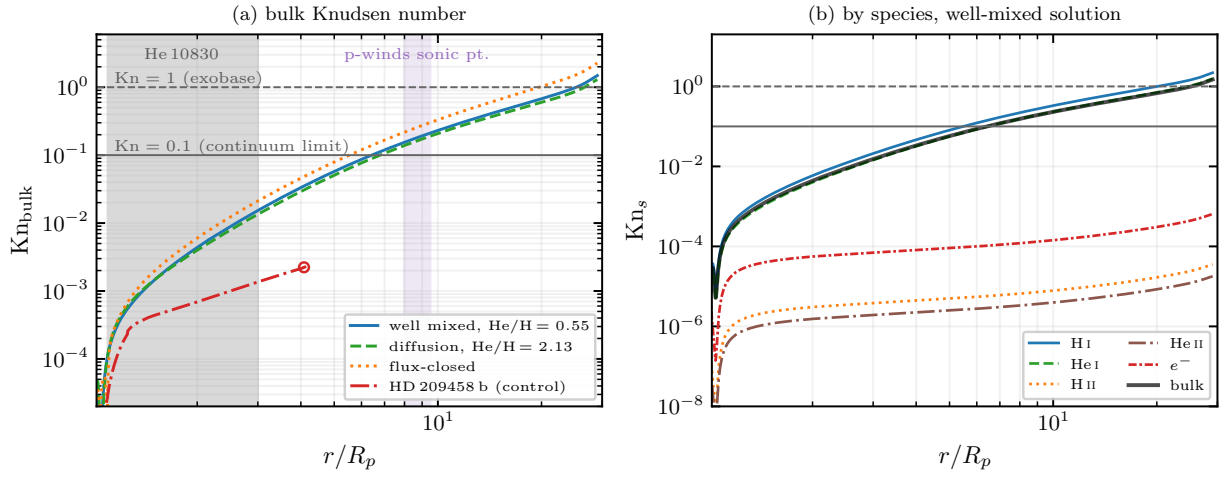


Figure 19: **Where the continuum description holds.** (a) The bulk Knudsen number of the three representative LHS 1140 b solutions and of the HD 209458 b control (open circle: its sonic point at $4.085 R_p$). Shaded: the He 10830 line-forming region, $1.1\text{--}3 R_p$, and the $8\text{--}9.5 R_p$ at which the p-winds retrieval places its sonic point. The LHS 1140 b solutions cross $Kn = 0.1$ at $5.5\text{--}6.8 R_p$ and $Kn = 1$ at $19.8\text{--}26.7 R_p$, both well inside the domain; the control stays below $Kn_{\text{bulk}} = 2.3 \times 10^{-3}$ everywhere and is collisional through its critical point, over which its largest Kn is 0.021 against $1.9\text{--}3.0$ for the LHS 1140 b solutions — a factor of about 100. (b) The species split of the well-mixed solution: neutral hydrogen and neutral helium set the number, while the ions and electrons are held by Coulomb collisions two to five orders of magnitude more tightly.

what a collisionless outer atmosphere at the same density and temperature would lose, so the failure of the assumption is not hiding an order of magnitude — it is hiding a factor of a few, in an undetermined direction. On HD 209458 b, where $\lambda_J = 10.85$ and the solution is validated through its critical point, the hydrodynamic rate exceeds Jeans by 10^4 : the signature of a driven wind rather than an evaporating atmosphere.

Which profile pair is read moves the exobase, not the verdict. On the equilibrium profiles instead of the advection-corrected ones the LHS 1140 b solutions are still subsonic across the domain ($\mathcal{M}_{\max} = 0.82$), still reach $\text{Kn} \sim 1$ where the outer boundary sets the flux, and the control is still collisional through its critical point. The exobase, however, moves above $30R_p$, because the equilibrium far field is more ionized and Coulomb collisions there are far stronger than neutral ones. An exobase radius quoted from this diagnostic has to state which pair it came from; the numbers above are the advection-corrected pair throughout.

12.3 What this changes, and what it leaves open

- **The outer boundary condition closes this wind, not a critical point.** For a transonic solution the critical point fixes the mass flux and the outer boundary is passive. Here it is the other way round, and the boundary sits in gas that is already collisionless. Any $\dot{M}$ quoted for LHS 1140 b from these solutions — including the $\log_{10} \dot{M} = 7.500$ of the flux closure (Sect. 11.) and the comparison with the rate the retrieval imposes — has to carry that warning. The same applies to the p-winds number, whose sonic point sits at $\text{Kn} = 0.14\text{--}0.30$.
- **The rate itself is not settled here, and cannot be.** What it would take is a kinetic or transitional-flow calculation — DSMC, or a 13-moment closure carried through the transitional region — and this repository contains neither. It is recorded as open, not estimated.
- **The line analysis stands.** The He 10830 layer is collisional by two orders of magnitude, so the equivalent-width matching, the composition inference conditional on it, and the invariant 22 km s^{-1} width deficit are unaffected by this section. It constrains what the escape rate may be quoted as, not what the line says.

13. Status against the goal

The line is not reproduced. What stands:

- **Amount of absorption: matched, and the composition it takes is a pair set by the lower boundary** — He/H = 2.09 above a prescribed metal-free scalar base, 10.7 above a lower atmosphere solved as a photochemical column and handed over as a profile (Sect. 11.6). Both carry H/He element diffusion at the same adopted $K_{zz} = 10^9 \text{ cm}^2 \text{ s}^{-1}$, so the base is what separates them, not the mixing. Inside the prescribed-base branch the answer is further conditional: 0.060 well mixed on the GJ 699 proxy, 0.55 well mixed on the GJ 1132 proxy, 4.68 wherever $K_{zz} \leq 10^5$ leaves the molecular coefficient alone. In each case the crossing is insensitive to the broadening treatment at the 2% level. The match is always available, so it constrains the composition only as far as the assumptions behind it are secure.

- **Line shape: not matched**, by a factor 3.4 in both depth and width, and not fixable by composition. The missing ingredient is 22.30 km s^{-1} FWHM of velocity spread ($\sigma = 9.5 \text{ km s}^{-1}$), measured in Sect. 8.2 and Fig. 4; supplied as one Gaussian it puts depth, blue depth and red/blue ratio inside the observed 1σ interval at once, but its physical origin is unidentified. The demand is unchanged under diffusion and the adopted K_{zz} : measured the same way on the EW-matched diffusive solution it is 22.31 km s^{-1} , 0.01 km s^{-1} away (Sect. 8.3); it is 22.39 km s^{-1} on the flux-closed solution (Sect. 11.); and it is 22.08 km s^{-1} on the $5.1\times$ brighter proxy spectrum (Sect. 10.). The adiabatic metastable bump, which would have supplied it without a free parameter, is present and measured (Sect. 8.5) but sits too slow to contribute.
- **Escape rate: an output, not a fit, and SED-set.** $\dot{M} \simeq 6 \times 10^7 \text{ g s}^{-1}$ on the GJ 1132 proxy, a third of the value the retrieval imposes, and nearly independent of composition. On the GJ 699 proxy, $5.1\times$ brighter in XUV, it is $4.0 \times 10^8 \text{ g s}^{-1}$ — a factor 7 higher and within 0.04 dex of the value the paper retrieves on that same proxy (Sect. 10.).
- **That escape rate is unvalidated in the collisional sense; the line is not.** All three representative solutions are subsonic to $29R_p$ ($\mathcal{M}_{\text{max}} = 0.54\text{--}0.57$), so the mass flux is set at the outer boundary, where $\text{Kn}_{\text{bulk}} = 1.3\text{--}2.3$; the exobase is at $19.8\text{--}26.7R_p$ and the p-winds sonic point at $8\text{--}9.5R_p$ sits at $\text{Kn} = 0.14\text{--}0.30$, so the critical region of this planet is transitional for both codes. The He 10830 layer at $1.1\text{--}3R_p$ is collisional by two orders of magnitude ($\text{Kn} \leq 0.021$), and an HD 209458 b control validates cleanly at a factor 100 lower Kn (Sect. 12.). What the true rate is needs a kinetic or transitional-flow calculation, which this repository does not contain.
- **Composition: base-limited and SED-limited, and weakly K_{zz} -limited.** In the well-mixed limit the equivalent-width crossing is $\text{He}/\text{H} = 0.550$ on one proxy spectrum and 0.060 on the other, helium-rich against helium-poor, with the measurement unable to separate them. With binary H/He diffusion active the crossing rises, and the whole K_{zz} axis has now been walked: $\text{He}/\text{H} = 4.68$ on the plateau at $K_{zz} \leq 10^5$, then 4.45, 3.78, 2.87, 2.09, 1.54 and 1.11 at 10^6 through $10^{11} \text{ cm}^2 \text{ s}^{-1}$ (Table 2, Fig. 5) — a factor 1.9 across the range the literature leaves open, 4.2 across the entire axis, against the factor 9 the SED carries. Those are all conditions inside the prescribed-base branch. Replacing that base with the solved photochemical column moves the crossing to 10.7 (Sect. 11.6), so end to end the inference spans 0.060 to 10.7, a factor 180. So any composition read off this line, the paper's included, is strongly conditional on the assumed XUV spectrum and on what is placed at the lower boundary, and weakly conditional on the eddy coefficient, as $K_{zz}^{-0.14}$ wherever the eddy term acts at all.
- **Composition as an output, on one axis, and the lower boundary raised to an axis of its own in its place.** With the lower atmosphere solved as a photochemical column and the elemental fluxes iterated to continuity across the $1 \mu\text{bar}$ match, He/H at the match is a solution: 2.09235, single-valued to 5.5×10^{-6} from trial fluxes a factor of ten apart, at $\log_{10} \dot{M} = 7.500$ (Sect. 11.). At the adopted $K_{zz} = 10^9 \text{ cm}^2 \text{ s}^{-1}$ that output is the well-mixed value — only 2×10^{-4} of it comes from the escape flux — so the conditionality moves onto K_{zz} rather than away. Above the match the wind still fractionates strongly, $F_{\text{He}}/F_{\text{H}} = 0.73$ by mass against the base reservoir's 8.31. The handoff also brings the column's C, N and O, which take $\log_{10} \dot{M}$ from 7.80 to 7.46 and the red-pair depth from 4.40% to 1.52%; **at that base the flux-closed solution at 2.09 no longer reproduces the equivalent width**, 0.421 against the measured $1.108 \pm 0.030 \text{ \% } \text{\AA}$.

- **The composition the line needs above a photochemical base:** $\text{He}/\text{H} = 10.7$. A ladder of seven flux-closed solutions, reservoirs 2.09 to 12.0 and each arm converged to its own fixed point, crosses the measured equivalent width between the 10.31 and 12.01 rungs: 10.7, with 10.5–10.9 for the measurement's 1σ interval, interpolated inside that bracket and not run at the crossing itself (Sect. 11.6, Table 9). The bracket is wide because the two rungs straddling the measurement have local power-law slopes of 0.25 and 1.42 against the 0.5–0.7 of the rest of the ladder, and that is not resolved. Two things follow: the closure condition alone does not select a reservoir here — every rung satisfies it, so the line selects, and what the closure adds is consistency with a photochemical lower atmosphere carrying its own C, N and O — and $\dot{M}$ is flat across the whole ladder, $\log_{10} \dot{M} = 7.39\text{--}7.50$, i.e. an eighth of the retrieval value at the crossing. Both numbers carry the width shortfall (FWHM 0.267 against 0.841 Å) and the collisional caveat above. This is the answer on the solved-column branch; 2.09 is the answer to the same question above a prescribed metal-free base, and neither is an independent prediction of the model.
- **Why the same helium absorbs differently over the two bases, measured.** Adding the column's C, N and O at fixed reservoir takes radiative cooling from 6–9% of the heating to 71% at $1.2R_p$ (95% of it metal-line cooling, largest single contributor neutral C I) and T_{max} from 5330 to 3944 K; the metastable peak rises and the column inside $2R_p$ is unchanged to 0.4%, but the column outside $2R_p$ falls by 2.6, which is the factor the equivalent width falls by. Raising the reservoir to 10.31 on that same base returns T_{max} to 5869 K — past the metal-free 5330 — with the mean mass per particle at $2R_p$ up by 1.68 and the heating per particle up by 1.88, and with the C/N/O density diluted along with the hydrogen it is referred to. In the metastable balance the peak ratio $50.2 \rightarrow 122.5$ splits as a source up by 1.55 against a lifetime up by 1.57, the latter because Penning ionization on H I — 89% of the sink at 2.09, 60% at 10.31 — slows as hydrogen is thinned (Sect. 11.5, Table 8, Fig. 16). *Read as an interpretation*, the ladder is a thermal trade between the metal content of the base and the helium content above it, and one line cannot separate them.
- **The 2025 non-detection sits at about 0.3 of the fiducial XUV.** Scaling the irradiation with the lower atmosphere held fixed, the red-pair depth falls below the 0.6 per cent limit of the 2025 epoch at $\simeq 0.30$ of fiducial in the scalar-base model and just under 0.33 in the flux-closed one (Sect. 11.7, Table 10) — about $10 \text{ erg cm}^{-2} \text{ s}^{-1}$ or less. The two models agree there despite differing by 4.6 in reservoir and 2.5 in escape rate; they do not agree on the slope with which they get there, which is a property of each model and not a result.
- **Not verified: the second-chemistry-code cross-check.** VULCAN did not converge on this cold helium-rich column — its convergence measure rose against the criterion with the timestep pinned, the limiting cell being H_2O at the cold trap — so no composition from a second code, and no cross-code spread, is quoted (Sect. 11.).
- **One solution is not converged to tolerance.** $\text{He}/\text{H} = 100$ stagnates at $\|R\| = 1.18 \times 10^{-3}$ from every seed tried; neighbours bracket it smoothly, so it is retained with the residual recorded.

Four axes have now been walked — the assumed XUV spectrum, the mixing assumption with its eddy coefficient, composition itself, and what is placed at the lower boundary — and each separates the two questions rather than answering them together. The last of them is the one that splits the answer in two rather than shifting it: 2.09 above a prescribed metal-free base, 10.7 above a solved photochemical column. Every one of them moves the composition

the equivalent width implies, two of them by a large factor; none of them moves the width. What is left as a possible source of the missing 22 km s^{-1} is therefore what it was: velocity fields that are three-dimensional, or heating a 1D hydrodynamic energy equation does not carry. The adiabatic-cooling metastable bump, the one candidate that would have needed no free parameter, has been looked for on both spectra and does not supply it for this host (Sect. 8.5).

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